

Chapter 2

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Introduction

- Software project management begins with a set of activities that are collectively called project planning.
- **Before the project start**, the software team should estimate:
 - ✓ the **work** to be done,
 - ✓ the **resources** required (e.g. ___) , and
 - ✓ the **time** required from start to finish
- Once these activities are accomplished, the team should **develop a project schedule** that
 - ✓ Defines **the software engineering tasks** and **milestones**,
 - ✓ Identifies **who** is responsible for conducting each task, and
 - ✓ Specifies the inter-task **dependencies**





2.1 Project Planning and Estimation

- Project **Planning**
- **Estimations** in Projects

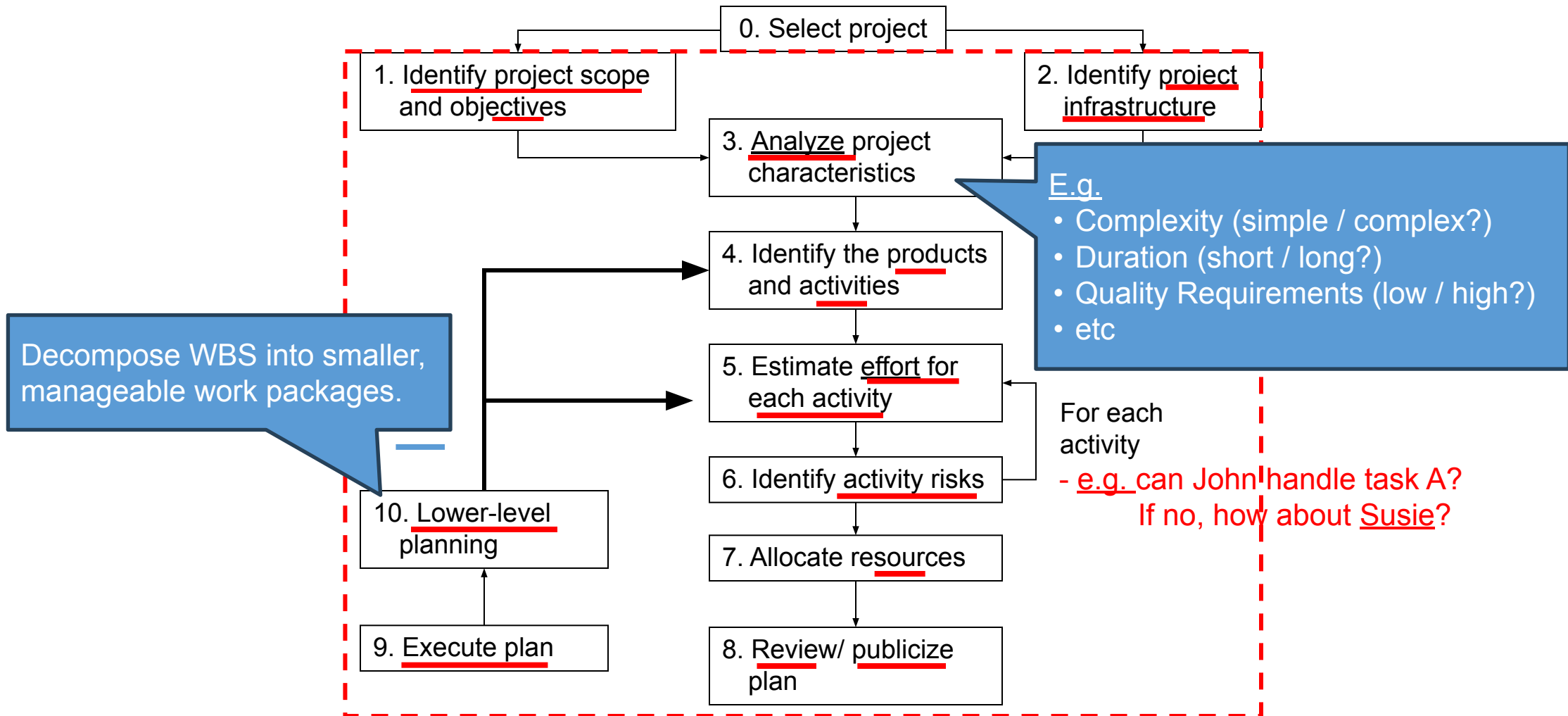
Project Management Process Group and Knowledge Area Mapping

	Initiating	Planning	Executing	M&C	Closing
Integration	Develop Project Charter	Develop PM Plan	Direct & Manage Project Work	- M&C Project Work - Perform Integrated Change Control	Close Project or Phase
Scope		<u>Plan</u> Scope Management - Collect Requirements - Define Scope - Create WBS		- Validate Scope - Control Scope	
Time		<u>Plan</u> Schedule Management - Define Activities - Sequence Activities - Estimate Activity Resources - Estimate Activity Durations - Develop Schedule		Control Schedule	
Cost		<u>Plan</u> Cost Management - Estimate Costs - Determine Budget		Control Costs	
Quality		<u>Plan</u> Quality Management	Perform Quality Assurance	Control Quality	
HR		<u>Plan</u> HR Management	- Acquire Project Team - Develop Project Team - Manage Project Team		
Communications		<u>Plan</u> Communications Management	Manage Communications	Control Communications	
Risk		<u>Plan</u> Risk Management - Identify Risks - Perform Qualitative Risk Analysis - Perform Quantitative Risk Analysis - Plan Risk Responses		Control risks	
Procurement		<u>Plan</u> Procurement Management	Conduct Procurements	Control Procurements	Close Procurements
Stakeholder	Identify Stakeholders	<u>Plan</u> Stakeholder Management	Manage Stakeholder Expectation	Control Stakeholder Engagement	

Project Planning

2.1 Project Planning and Estimation

Steps in Project Planning



Project Plan Content

Project Planning

Content of a project plan:

- a. **Introduction**
- b. **Background**: including reference to **the business case**
- c. **Project objectives**
- d. **Constraints** – these could be included with project objectives
- e. **Method**
- f. **Project products**: both deliverable products that the **client will receive** and intermediate products
- g. **Activities** to be carried out
- h. **Resources** to be used
- i. **Risks** to the project
- j. **Management of the project**, including
 - Organizational responsibilities
 - Management of quality
 - Configuration management



Stakeholders

Project Planning

- **People who have a stake or interest in the project.**
- **Can be categorized as:**
 - **The project team** – they are directly involved in doing the project work.
 - **External to the project team but within the same organization.** E.g. the project leader might need the assistance of the users to carry out system testing
 - **External to both the project team and the organization.**
Examples:
 - **contractors** who will carry out specific works for the project.
 - **suppliers** who supplies hardware
 - **customers** (or users) who will benefit from the system that the project implements
- Different types of stakeholders may have different interests. Thus, the **project leader must recognize** these different **interests** and to be able to reconcile them.



Setting Project Objectives

Project Planning

- Among the stakeholders are those who actually own the project and control the financing of the project. Hence, they also set the objectives of the project.
- The **objectives should define what the project team must achieve for project success.**
- **Objectives focus on the desired outcomes of the project** rather than the tasks within it – they are the post-conditions of the project.



Setting Project Objectives

Project Planning

- In order to achieve the goal, we must achieve certain objectives first.
 - i.e., the objectives are steps on the way to achieve a goal.
- An effective objective for an individual must be something that is within the control of that individual.
 - e.g., an appropriate objective for software developers would be to keep development costs within a certain budget.



SMART

Project Planning

Well-defined objective should be:

- **Specific** – The project objective should be clear, well-defined, and unambiguous.
Obj = To achieve minimum customer satisfaction rating of 8/10 by 30-Jun.
This obj is specific. It clearly states the desired outcome.
- **Measurable** – there should be measures of effectiveness which tell us how successful is the project.
Customer satisfaction is measurable right?
- **Achievable** – it must be within the power of the individual or group to achieve the objective
* Cust satisfaction target = 8/10: achievable?
* If target = 10/10: achievable?
- **Relevant** – the **objective** must be relevant to the true purpose of the **project**.
- **Project** is “something” that you will do/carried out
- **Objective** is the “desired outcome” that you want to achieve
- **Time constrained** – there should be a defined **point in time** by which the objective should have been achieved.
30-Jun is the deadline to achieve the desired outcome

Obj = “To achieve higher level of customer satisfaction.”
Does this obj clearly states the desired outcome? **No**

Example 1:

To achieve minimum customer satisfaction rating of 8/10 by 30-Jun through training customer service team

Ref: Appendix 2.1
Taxonomy of Project Definitions

Example 2:

(this is not a SMART obj)

To develop a Hotel Room Booking System

which allowed users to complete their job which **save a lot of time** and **they will like it.**

(source: FYP proposal)

SMART

This obj did not mention the amount of time it will help to saved

Well-defined objectives should be:

- **Specific** –project objective should be clear, well-defined, and unambiguous.
(Obj= **reduce booking time to 10 mins at a cost not more than \$100,000 by 01-05-2026.** This obj is **specific**. It clearly state the desired outcomes
- **Measurable** – there should be measures of effectiveness which tell us how successful is the project.
1) Amount of time (months) taken to complete the proj
3) Amount money spent to complete the project
2) Time (minutes) user spent to complete room booking
- **Achievable** – it must be within the power of the individual or group to achieve the objective.
- **Relevant** – the objective must be relevant to the true purpose of the project.
- Project is "something" that you will do/carried out
- Objective is the "desired outcome" that you want to achieve
- **Time constrained** – there should be a defined point in time by which the objective should have been achieved.

This project is considered successful if:
1) Completed not later than **01-05-2026**
2) Development cost **<= 100K**
3) Time to complete booking is **<= 10'**

Improved Version

To implement a Hotel Room Reservation System

by **01-05-2026**

that will reduce booking time to **10 mins** at a cost not more than **\$100,000.**



Setting Project Objectives

Project Planning

- Among the stakeholders are those who actually own the project and control the financing of the project. Hence, they also set the objectives of the project.
- The **objectives should define what the project team must achieve for project success.**
- **E.g. “To implement a Hotel Room Reservation System by 01-05-2026 that will reduce booking time to 10 mins at a cost not more than \$100,000”**

Written in this way allow us to check how successful is the project team



Gantt Chart

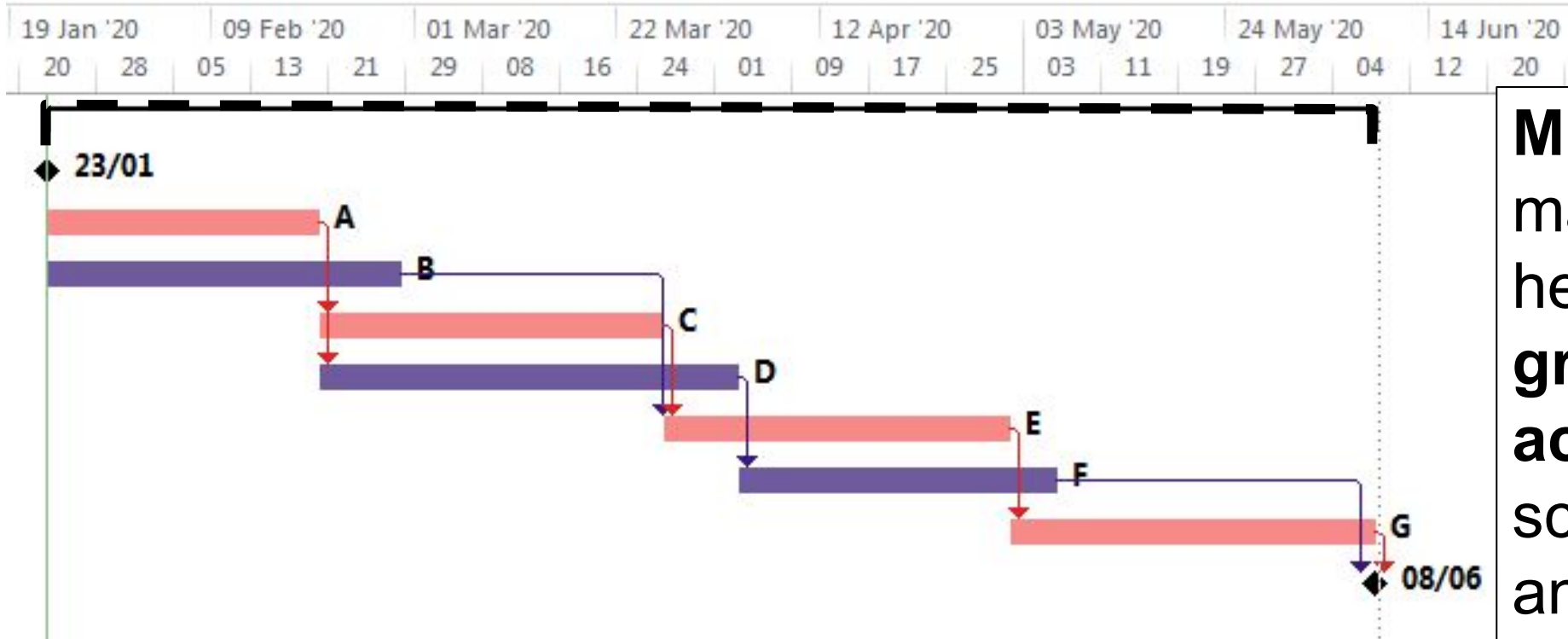
Project Planning

- A bar chart where all **project tasks** are listed in the left-hand column and the **horizontal bars indicate the duration of each task**.
- When multiple bars occur at the same time on the calendar, task concurrency is implied.
- The **diamonds indicate milestones**.
- Produces project tables (i.e. a tabular listing of all project tasks, their planned and actual start and end dates) which enable you to track progress.



Gantt Chart

Project Planning



Milestone (◇) – a marker use to help in identifying **group of activities**, set schedule goals and monitor progress

Gantt Chart

Project Planning

— Project Schedule / Project Table

Task Mode ▾	Task Name ▾	Duration ▾	Start ▾	Finish ▾	Predecessors ▾
1	Project Z	97.55 days	Thu 23/01/20	Mon 08/06/20	
2	Start	0 days	Thu 23/01/20	Thu 23/01/20	
3	A	4 wks	Thu 23/01/20	Wed 19/02/20	
4	B	5.33 wks	Thu 23/01/20	Fri 28/02/20	
5	C	5.17 wks	Thu 20/02/20	Thu 26/03/20	3
6	D	6.33 wks	Thu 20/02/20	Fri 03/04/20	3
7	E	5.17 wks	Thu 26/03/20	Fri 01/05/20	4,5
8	F	4.5 wks	Fri 03/04/20	Wed 06/05/20	6
9	G	5.17 wks	Fri 01/05/20	Mon 08/06/20	7
10	Finish	0 days	Mon 08/06/20	Mon 08/06/20	8,9

define the dependencies between tasks

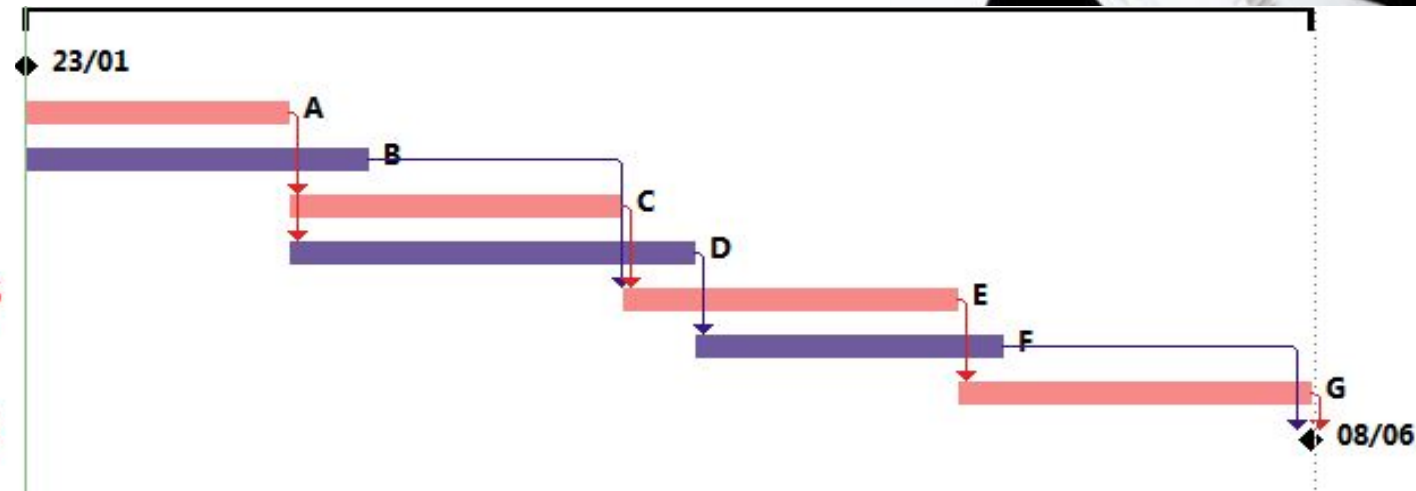
Task no 3 must be completed before task no 5 can start

Task no 4 & 5 must be completed before task no ____ can start

Project Schedule + Gantt Chart

Project Planning

Task Mode	Task Name	Duration	Start	Finish	Predecessors
1	Project Z	97.55 days	Thu 23/01/20	Mon 08/06/20	
2	Start	0 days	Thu 23/01/20	Thu 23/01/20	
3	A	4 wks	Thu 23/01/20	Wed 19/02/20	
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7	E	5.17 wks	Thu 26/03/20	Fri 01/05/20	4,5 4,5
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Program Evaluation and Review Technique (**PERT**)

Project Planning

Project Schedule					
Task Mode	Task Name	Duration	Start	Finish	Predecessors
1	Project Z	97.55 days	Thu 23/01/20	Mon 08/06/20	
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Use Program Evaluation and Review Technique (**PERT**) to develop a better time estimate for each task

Q) How to develop a better time estimates for each task/activity? OR How to develop a better time estimates when there is a high degree of uncertainty about the individual activity duration estimates?

Ans: Use PERT (~~aka network diagram~~)

PERT uses 3 estimates for the duration of each task instead of one!

- Optimistic time
- Most likely time
- Pessimistic time

Program Evaluation and Review Technique (PERT)

Project Planning

PERT Step 1: Identify specific activities and milestones

Activity
Start
A
B
C
D
E
F
G
Finish



Program Evaluation and Review Technique (PERT)

Project Planning

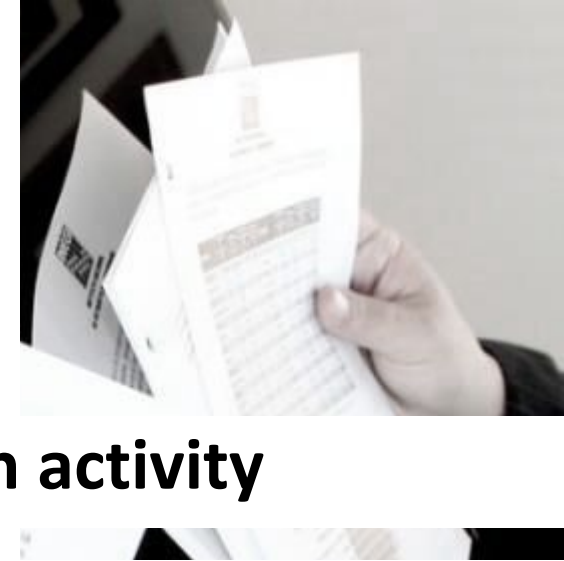
PERT Step 2: Specify Predecessor to sequence project activities

Activity	Predecessor
Start	
A	-
B	-
C	A
D	A
E	B, C
F	D
G	E
Finish	D, E



Program Evaluation and Review Technique (PERT)

Project Planning



PERT Step 3: Estimate the time for each activity

Activity	Predecessor
Start	
A	-
B	-
C	A
D	A
E	B, C
F	D
G	E
Finish	D, E

- Use the 3-time estimates:
 - **Optimistic time (O):** the **shortest time** in which the activity can be completed
 - **Most likely time (ML):** the **completion time that has the highest probability**
 - **Pessimistic time (P):** the **longest time** in which the activity can be completed.
- Then, **calculate the expected time** using the weighted average formula.

$$t_e = (O + 4*ML + P) / 6$$

Program Evaluation and Review Technique (PERT)

Project Planning

PERT Step 3: Use 3-time est. to est. the time for each activity
calculate the Expected Time (t_e) for each activity

Activity	Predecessor	Activity Durations (Weeks)			
		O	ML	P	Expected (t_e)
Start					
A	-	2	4	6	4
B	-	3	5	9	5.33
C	A	4	5	7	5.17
D	A	4	6	10	6.33
E	B, C	4	5	7	5.17
F	D	3	4	8	4.5
G	E	3	5	8	5.17
Finish	F, G				

4.5 weeks is a BETTER time estimate for activity F

Formula

$$(O + 4*ML + P) / 6$$

$$(3 + 4*4 + 8) / 6 = 4.5$$

For task/activity F:

Using PERT = 4.5 weeks ☐ 5 weeks

- **Without** using PERT = weeks.
 (based on most likely(ML))

Can be completed in 3 weeks instead of 4

Most likely need 4 weeks

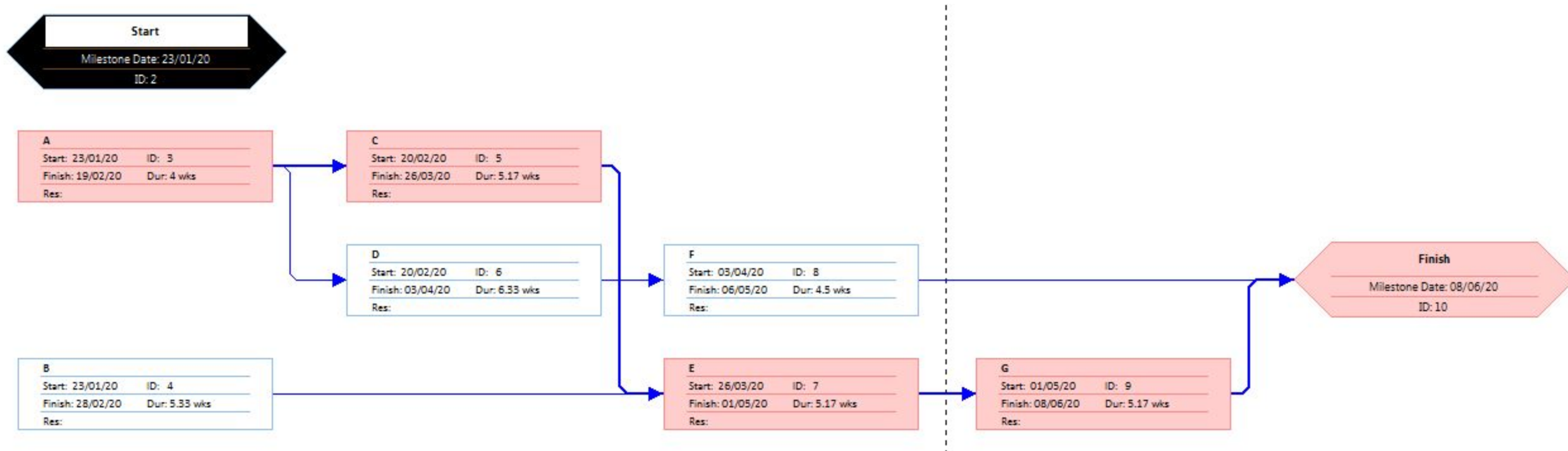
4 weeks is not possible! 8 weeks instead

Program Evaluation and Review Technique (PERT)

Project Planning

PERT Step 4: Construct a network diagram

- You can represent both sequential and parallel activities in the diagram.
- Each activity represents a node; arrows to show relationships between activities.

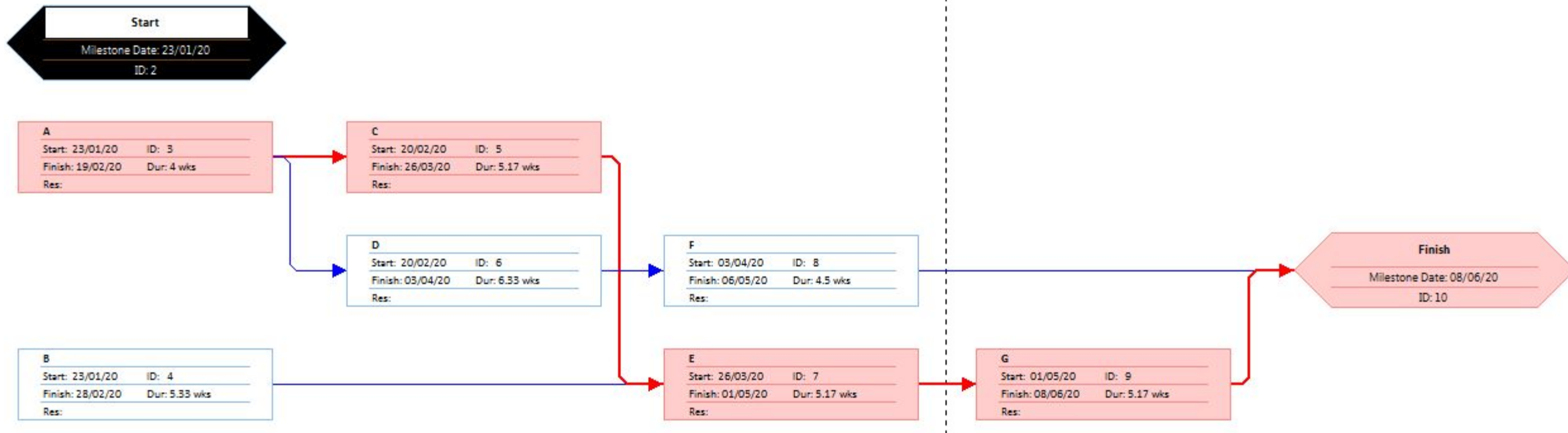


Program Evaluation and Review Technique (PERT)

Project Planning

PERT Step 5: Identify the critical path

- It's the **longest path** in the network diagram
- Any task on this path is **delayed**, it will **delay** the entire project.
- Identifying the critical path is crucial for project managers as it helps in scheduling, resource allocation, and ensuring the timely completion of a project.



Program Evaluation and Review Technique (PERT)

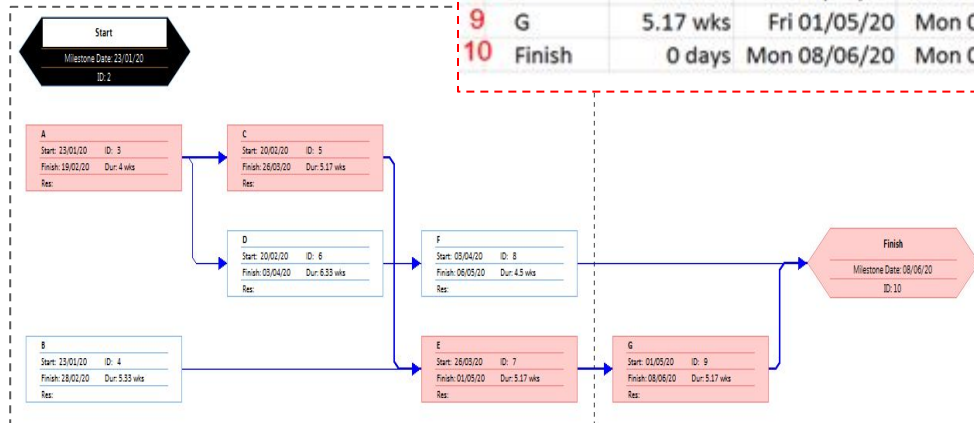
Project Planning

Below is a Network Diagram generated by Microsoft Project

Exercise: Draw this network diagram MANUALLY!

1. Click Task
2. Click Gantt Chart
3. Select Network Diagram

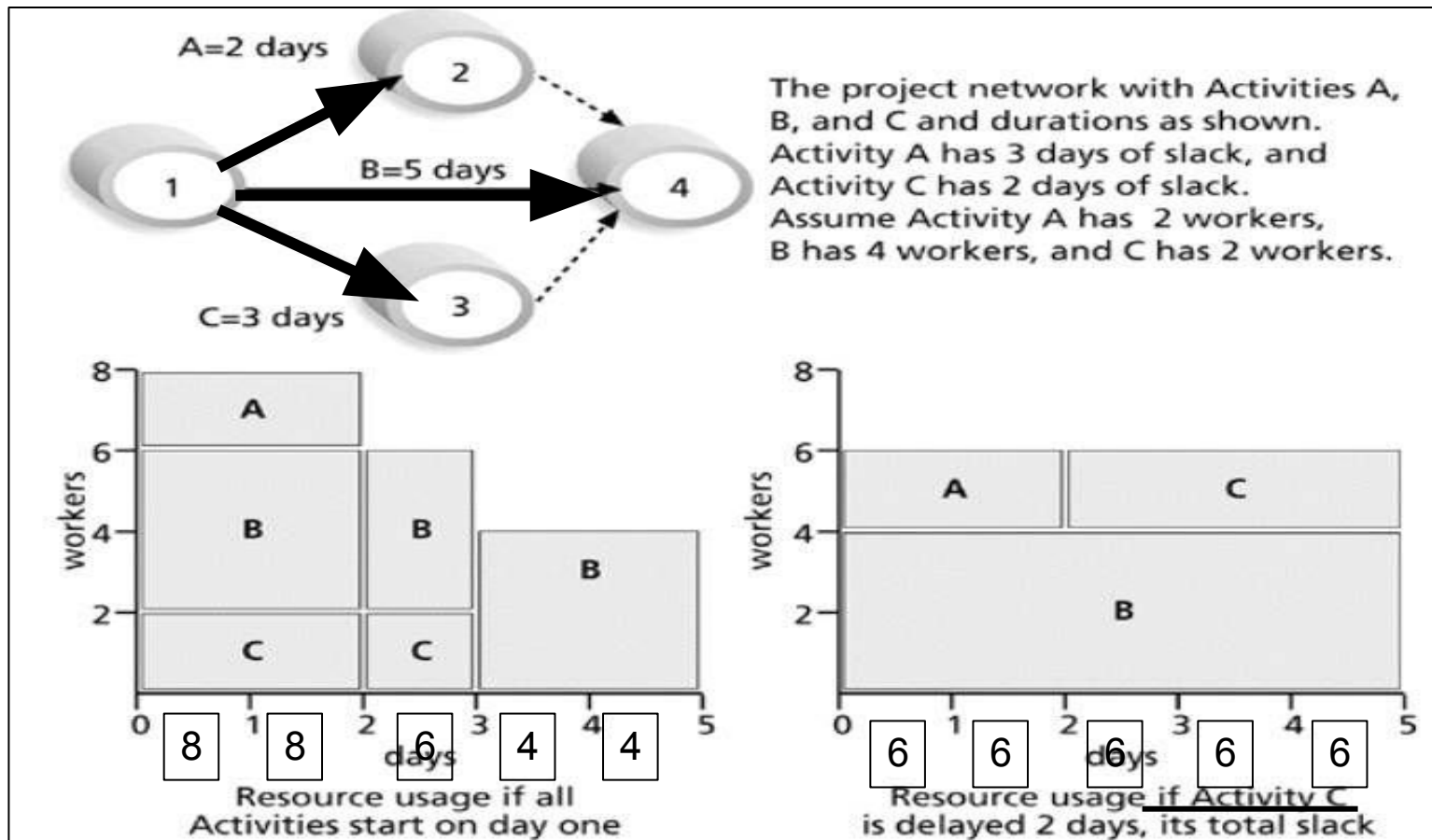
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Program Evaluation and Review Technique (PERT)

Project Planning

A PERT chart illustrates a project as a network diagram



Estimations in Projects

Project Planning

- Estimating **resources, cost and schedule** for a software project requires:
 - experience
 - access to good historical information (metrics) and
 - quantitative predictions.
- Estimation carries inherent risks, and this risk leads to uncertainty.
- Uncertainty in project planning:
 - **Project complexity**
 - **Project size**
 - Degree of structural **uncertainty** (i.e. the degree to which **requirements** have been solidified, the ease with which functions can be compartmentalized, and the hierarchical nature of the information that must be processed)



Estimations in Projects

Project Planning

To achieve reliable cost and effort estimates:

- **Delay** estimation until late in the project
- Base estimates on **similar projects** that have already been completed
- Use relatively simple decomposition techniques to generate project cost and effort estimates
- Use one or more empirical models for software cost and effort estimation



Estimations in Projects

Project Planning

Three-point Estimates *(for estimating task/activity duration)*

1. **Simple** Average
 - $(P+O+ML)/3$
 - Triangular distribution
2. **Weighted** Average
 - $(P+O+4ML)/6$
 - Beta distribution



Estimations in Projects

Project Planning

Simple Avg
 $(P+O+ML)/3$

Weighted Avg
 $(P+O+4ML)/6$

Activity	Optimistic	Pessimistic	Most Likely	Triangular Distribution	Beta Distribution
A	2	6	4	4	4
B	4	12	8	8	8
C	4	10	7	7	7
D	10	22	16	16	16
E	8	18	13	13	13
F	6	16	11	11	11
G	16	26	22	22	22
H	6	12	10	10	10

$$[6+12+(4*10)]/6 = 9.67$$

$$(6+12+10)/3 = 9.33$$

Round UP to the nearest whole number

Ch2: Project Planning, Control & Process Models Selection

Quick Review



2.1 Project Planning and Estimation

- ☐ Create a WBS (WBS is a list of activities to be carried out)
- ☐ Estimate the Start Date & End Date for each task/activity (duration=ED - SD)
- ☐ Set relationship between tasks ☐ Project Schedule(PS)
(specify the predecessor) *(PS allows you to view all tasks and the relationship between tasks)*
- ☐ Use **3-POINT ESTIMATE** to develop a BETTER time allocation for each task/activity instead of using a single estimate.
- ☐ Draw Network diagram ☐ To identify critical path

2.2 Project Monitoring and Control

2.3 Selection of Process Models





2.2 Project Monitoring and Control

- Tracking the **Schedule**
- **Cost** Monitoring
- Project Control - **Corrective Actions**
- **Earned Value Analysis (EVA)**

Project Management Process Group and Knowledge Area Mapping

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Stakeholder	Identify Stakeholders	Plan Stakeholder Management	Manage Stakeholder Expectation	<u>Control Stakeholder Engagement</u>	



Tracking the Schedule

2.2 Project Monitoring and Control

Tracking can be accomplished in a number of different ways:

- **Conduct periodic project status meetings** (each team member reports progress and problems)
- **Compare the actual start date against the planned start date for each project task** listed in the project table ...
- **Check whether formal project milestones have been accomplished by the scheduled date**
- **Evaluate the results of all reviews conducted** throughout the software engineering process
- **Use Earned Value Analysis** technique to assess progress quantitatively
- Meet informally with practitioners to obtain their subjective assessment of progress to date and problems on the horizon



Project Control - Corrective Actions

2.2 Project Monitoring and Control

- Project manager needs to monitor project progress in terms of the **time, cost and quality** (triple constraints)
- Control is more than monitoring and finding out problems, **take corrective actions** whenever and wherever needed.



Project Control - Corrective Actions

2.2 Project Monitoring and Control

- a. Adding more staff
- b. Adding different skills
- c. Reassigning tasks
- d. Increasing/decreasing individual supervision
- e. Improving methods of working
- f. Streamlining the process
- g. Changing resource priorities
- h. Re-planning the project
 - i. Changing the phasing of deliverables
 - j. Increasing/decreasing the number of inspection
- k. Encouraging the team
- l. Introducing incentives
- m. Subcontracting part of the work
- n. Negotiating changes in the specification

Project Control - Corrective Actions

2.2 Project Monitoring and Control

a. Adding more staff

One of the possibilities **when behind schedule**

- **Will work if a task can be partitioned**
(E.g. **Interview users** to gather requirements)
- **Drawback**
 - *Will incur additional cost*
 - *May further delay the task because need to consider communication, learning curve, time to get-up to speed & etc.*



Project Control - Corrective Actions

2.2 Project Monitoring and Control

b. Adding different skills

- An alternative to adding more staff, consider adding people with different or greater skills
- When problems need skill/knowledge or staff lack experience
 - Adding experienced or skilled staff to a team may yield results e.g Adding a person who is skillful in UI design



Project Control - Corrective Actions

2.2 Project Monitoring and Control

c. Reassigning Task

- No need to add staffs or skills
- **Simply switch tasks around** as each member has different strength
- May result in higher productivity or better quality of work as some are creative, some are thorough, while some are analytical
- Can make use of the knowledge of characteristics of the team members to assign tasks that will make use of their strength



Project Control - Corrective Actions

2.2 Project Monitoring and Control

d. Increasing individual supervision

- Problems with the work of members are known only when they deliver products
- Partitioning tasks and creating smaller deliverables to **exercise quality control more frequently**
- **To allow inexperienced/ less confident staff to obtain guidance**



Project Control - Corrective Actions

2.2 Project Monitoring and Control

e. Decreasing individual supervision

- Opposite when dealing with **experienced staff** who may resent too frequent check up
- May affect personal interest and affect the work quality
- Giving individual responsibility and increase job interest and motivation for larger deliverables
- Reduces work of supervision and re-channel resourceful **focus**



Project Control - Corrective Actions

2.2 Project Monitoring and Control

f. Improving methods of working

- Besides the work itself, need to consider the suitability of methods for the tasks
- E.g. using Joint Application Development (JAD) approach to reconcile what seemed to be mutually exclusive requirements between different users, staging workshop where all users can come together to thrash out the differences during analysis stage
- Using 4th GL to obtain users interface requirements



Project Control - Corrective Actions

2.2 Project Monitoring and Control

g. Streamlining the process

(Make the process more efficient **or** simplified the process **or** remove unnecessary activities)

- Some tasks like quality control may be bureaucratic and time-consuming
- Will be aggravated if the procedures or processes are vague
- Streamline processes by removing unnecessary activities and using standardized forms and process (**e.g. using standardised change request(CR) form or download CR form from the intranet**)



Project Control - Corrective Actions

2.2 Project Monitoring and Control

h. Changing resource priorities

- Access to some important resources may be limited and created bottlenecks
- Need to negotiate for better access or find alternative environment/facilities in which work can proceed
- **Examine the project critical path to relocate the priorities of access**



Project Control - Corrective Actions

2.2 Project Monitoring and Control

i. Re-planning the project

- Problems evaluation may show some fundamental flaws in the way project has been planned
- Although embarrassing, it is critical, project manager has to **rework the plan**
- Some tasks dependencies become clear after the work has started
- Need to note while new plans remove problems of old plan may introduce new risks
- A re-appraisal of the risks should be part of the re-planning process



Project Control - Corrective Actions

2.2 Project Monitoring and Control

j. Changing the phasing of deliverables

- Short of a complete revision of plan, it may prove effective by changing the phasing of the deliverables
- Planning may show as one end-product but analysis may reveal that the product can be partitioned into several discrete elements
- Some discrete elements may have higher priorities and may be possible to **consider phased delivery** to concentrate on more urgent requirements
- May also consider parallel working e.g. design overlap with analysis
- May introduce new risks and proper appraisal is needed



Project Control - Corrective Actions

2.2 Project Monitoring and Control

k. Decreasing the number of inspections

- **Only IF inspections (assessment) are uncovering an acceptably low no. of defects**
- Else problems will arise later and doesn't result in any positive effects



Project Control - Corrective Actions

2.2 Project Monitoring and Control

I. Increasing the number of inspections

(This is to solve quality problems)

- For critical systems and IF intermediate deliverables are less satisfactory
- Or there is a high level of defects on completed deliverables
- To ensure errors are discovered earlier and rectified more quickly
- **May delay the project when no. of inspections increases**



Project Control - Corrective Actions

2.2 Project Monitoring and Control

m. Encouraging the team

- Project fatigue may arise (lower productivity, increased absenteeism, resignations, complaints, etc)
- Actions may include:
 - refocusing on team's achievements to rekindle enthusiasm
 - organize social events in company time to engender a sense of team spirit
 - redistribution of work to provide development opportunities
 - a team building exercise of some sort
 - reducing the size of deliverables (at the end may be larger!)



Project Control - Corrective Actions

2.2 Project Monitoring and Control

n. Introducing incentives

- Depending on the organization, the project manager may or may not be able to offer financial incentives
- But incentives may be in other form, **e.g. time off, recognition, etc.**
- For multiple team, inter-team competition may be considered but practiced with care



Project Control - Corrective Actions

2.2 Project Monitoring and Control

o. Subcontracting part of the work

- Despite all the tools and techniques, **project** may **still not on schedule**
- Project manager may consider sub-contracting out the tasks to those with special skills/ facilities
- But responsibility will still be with the PM
- **Must make sure the contractors work to the required standards**



Project Control - Corrective Actions

2.2 Project Monitoring and Control

p. Negotiate changes to the specification

- **When all else fail**, may negotiate to change the specification (may be the original objective is too ambitious for the time or money)
- **Alternatively, may resort to phased delivery** with major functions being delivered first (**Not usually well received** but better than to fail the project totally)





Project Control - **Corrective Actions**

2.2 Project Monitoring and Control

- a. Adding more staff
- b. Adding different skills
- c. Reassigning tasks
- d. Increasing/decreasing individual supervision
- e. Improving methods of working
- f. Streamlining the process
- g. Changing resource priorities
- h. Re-planning the project
- i. Changing the phasing of deliverables
- j. Increasing/decreasing the number of inspection
- k. Encouraging the team
- l. Introducing incentives
- m. Subcontracting part of the work
- n. Negotiating changes in the specification

**What else
can be
added here?**

Previous lesson

2.1 Project Planning and Estimation

Project Planning

Before the project start, estimate:

- ✓ the **work** to be done,
- ✓ the **resources** required (e.g. ___) , and
- ✓ the **time** required from start to finish

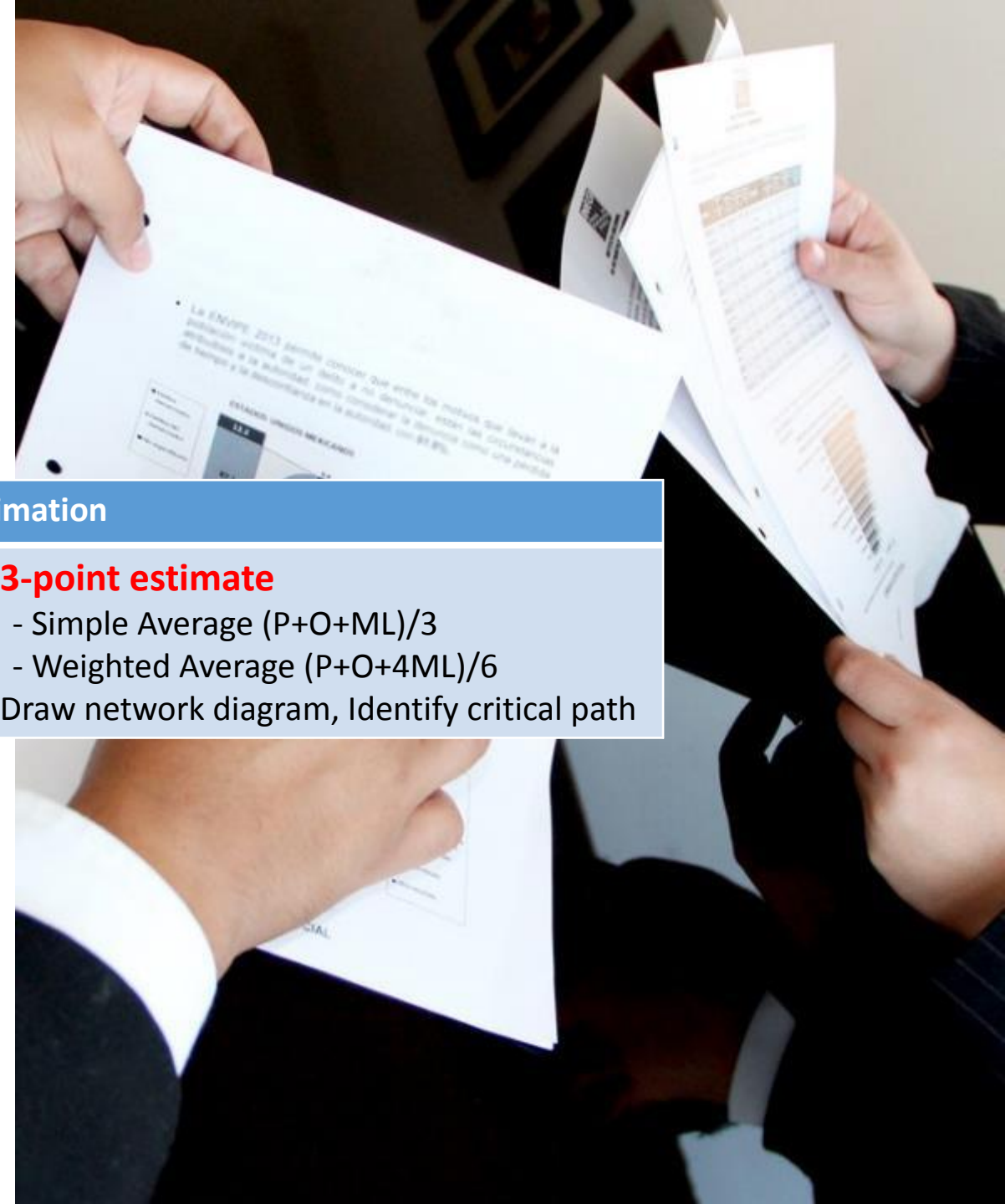
2.2 P

- Tracking the Schedule
- Project Control – **Take Corrective Actions** if project is behind schedule

2.3 Selection of Process Models

Estimation

- **3-point estimate**
 - Simple Average $(P+O+ML)/3$
 - Weighted Average $(P+O+4ML)/6$
- Draw network diagram, Identify critical path





Monitoring Cost

2.2 Project Monitoring and Control

- Cost/expenditure monitoring provides an indication of the effort that has gone into (or at least been charged to) a project.
- A project might be on time but only because more money has been spent on activities than original budgeted.
- A cumulative expenditure chart provides a simple method to **compare planned costs against actual costs**.
 - By itself it is not particularly meaningful, e.g. the expenditure chart could illustrate a project that is running late or one that is on time but has shown substantial cost savings.
 - Thus, we need to take into account the current status of the project activities before attempting to interpret the meaning of recorded expenditure.



Monitoring Cost (Compare planned cost against actual cost)

2.2 Project Monitoring and Control

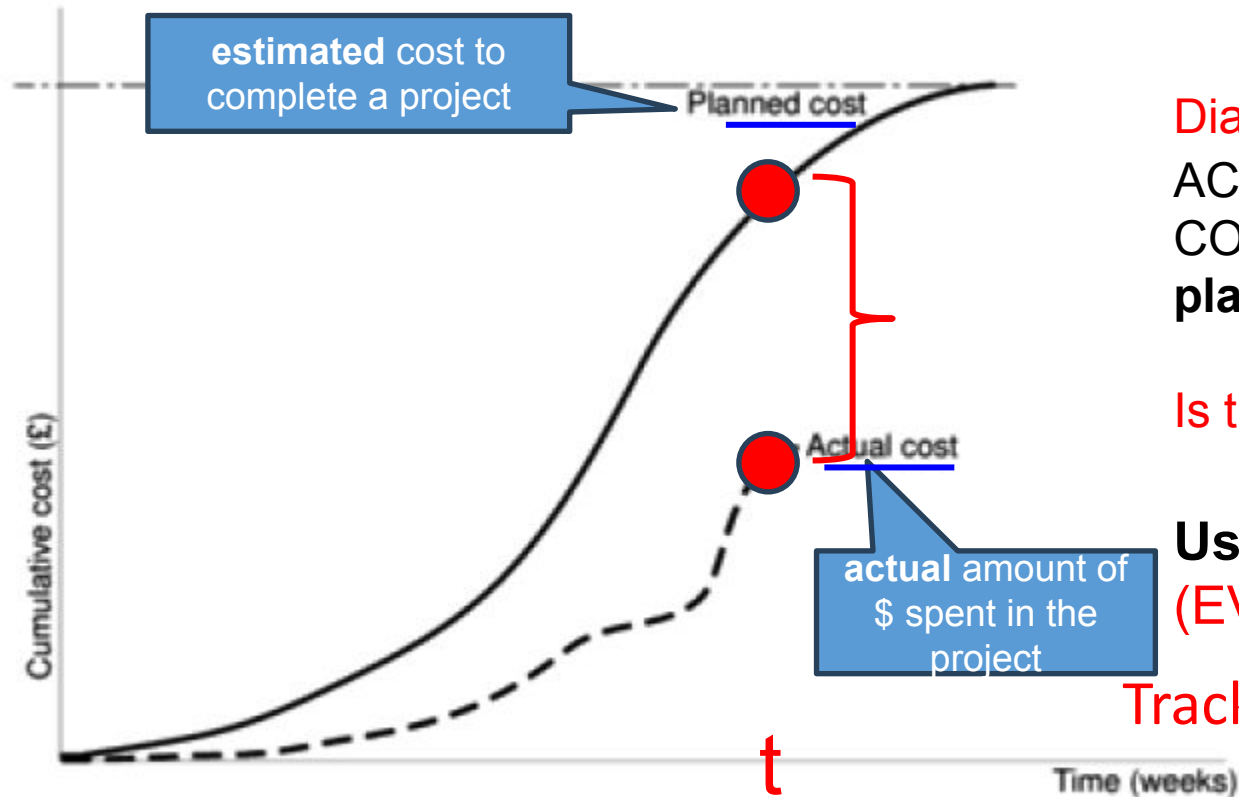


Diagram shows, at time t ,
ACTUAL COST < PLANNED
COST (i.e. **actual** amount paid <
planned amount to be paid)

Is there substantial costs savings?

Use **Earned Value Analysis (EVA)** to draw conclusion

Track **cumulative** expenditure

Earned Value (EV) Analysis

2.2 Project Monitoring and Control

- EV analysis is a refinement of cost monitoring.
(takes into account the project progress and schedule)
- **After doing EV analysis you will be able to check whether:**
 - 1. project **cost** has exceeded its **budget**
 - 2. project is **behind schedule** or **ahead of schedule**
- **EV Analysis** is based on assigning a “value” to each task (as identified in the WBS) based on the original expenditure forecasts. **This assigned value is called *planned value (PV)*.**
- **PV = the price agreed upon by a contractor to do the unit of work OR value of the work planned to be completed**
- **EV = the actual value of work completed**
- A task that has not started, the **EV = 0**,
EV = PV when it has been completed ...

Earned Value (EV) Analysis

2.2 Project Monitoring and Control

5 Techniques for Crediting or assigning EV to a Project

- **0/100 technique:** a task is assigned a value of 0 until it is completed, at which it is given a value of 100% of the budgeted value. (EV=0 when the job starts, when the job is finished EV=PV)
- **50/50 technique:** a task is assigned a value of 50% of its value as soon as it is started and then given a value of 100% once it's completed. (EV=50% of the PV when the job starts, when the job is finished EV=PV)
- **75/25 technique:** a task is assigned a value of 75% of its value as soon as it is started and then given a value of 25% once it's completed. (EV=75% of the PV when the job starts, when the job is finished EV=PV)
- **milestone technique:** a task is given a value based on the **achievement of milestones that have been assigned values** as part of the original budget plan. (assign a value to EV based on milestone achievement)
- **% complete:** the EV is based on the % of the project that has been completed. (assign a value to EV based on percentage of work completed)

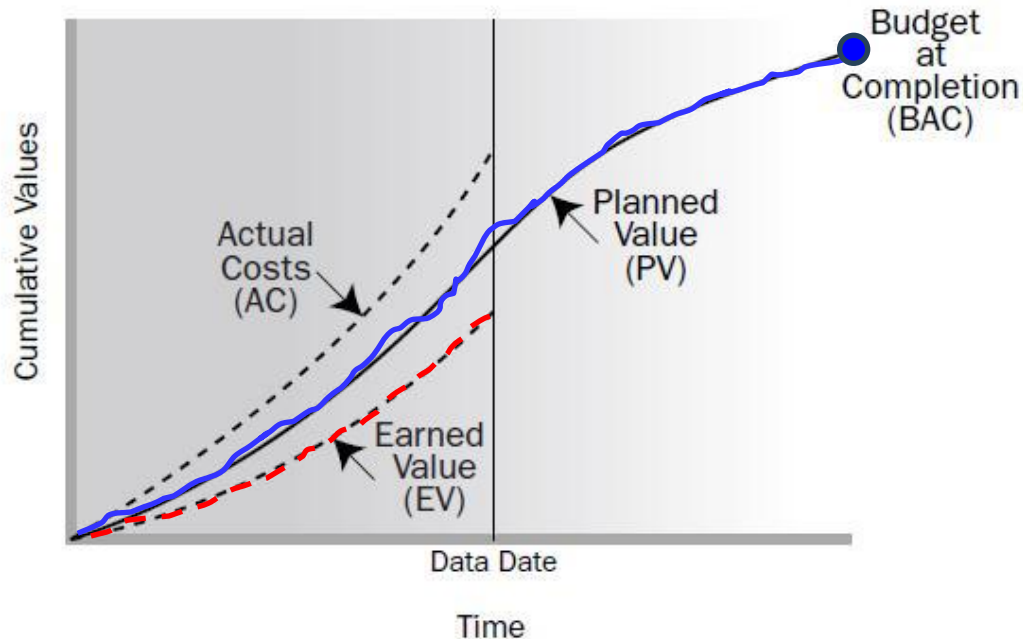
Step 1: Create the baseline budget

- The baseline budget (i.e. planned value, PV) is created based on the project plan.
- Shows the forecast growth in earned value through time.
- May be measured in monetary values or person-hours (or workdays).
- This example uses the 0/100 technique for crediting EV to the project.

Task ID	Task	Predecessor	Budgeted workdays	Scheduled Completion	Cumulative workdays	% cumulative earned value
1	Specify overall system	-	34	34	34	14.35
2	Specify module B	1	15	49	49	20.68
3	Specify module D	1	15	49	64	27.00
4	Specify module A	1	20	54	84	35.44
5	Check specifications	4	2	56	86	36.29
6	Design module D	5	4	60	90	37.97
7	Design module A	5	7	63	97	40.93
8	Design module B	6	6	66	103	43.46
9	Specify module C	3	25	74	128	54.01
10	Check module C spec	9	1	75	129	54.43
11	Design module C	10	4	79	133	56.12
12	Code and test module D	6	25	85	158	66.67
13	Code and test module A	7	30	93	188	79.32
14	Code and test module B	8	28	94	216	91.14
15	Code and test module C	11	15	94	231	97.47
16	System integration	15	6	100	237	100.00

Step 2: Monitor the earned value

- Having created the baseline budget, the next task is to monitor earned value (EV) as the project progresses.
- This is done by monitoring the completion of tasks (or activity starts and milestone achievements in the case of other crediting techniques).
- In addition, the actual cost (AC) of each task is also recorded.

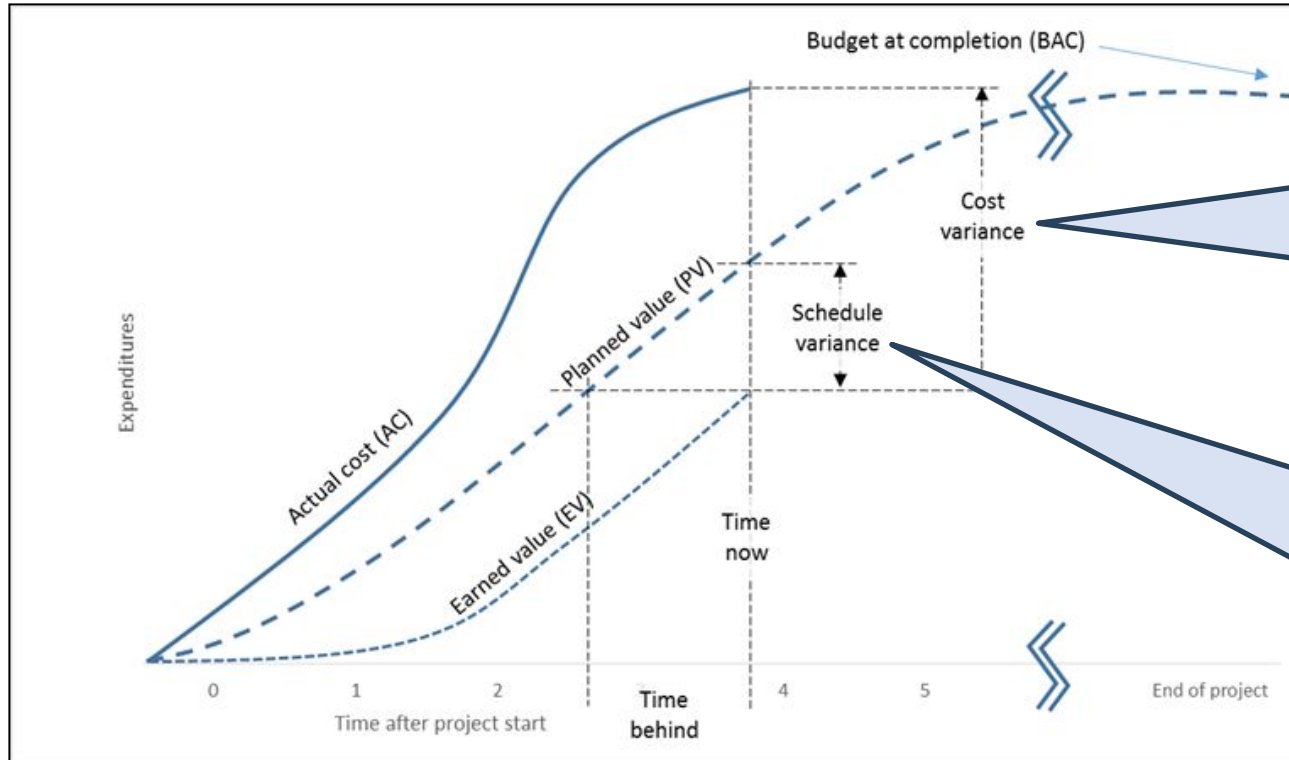


Earned Value Management (EVM) – Basic Metrics

- ✓ **Planned Value (PV):** Value of the work planned to be completed
- ✓ **Budget at Completion (BAC):** Budget for the whole project
- ✓ **Earned Value (EV):** Actual value of the work completed (*5 ways to assign a value to EV*)
- ✓ **Actual Cost (AC):** Actual cost incurred for the work completed

Step 3: Calculate Performance Statistics

(after knowing PV and EV)



Cost Variance (CV)

$$= EV - AC$$

- **negative** indicates cost has exceeded its budget
- **positive** means ...

Schedule Variance (SV)

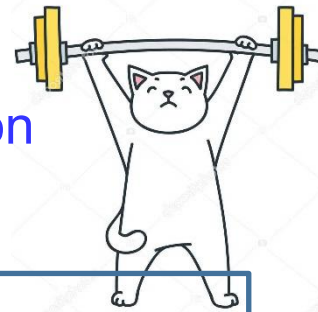
$$= EV - PV$$

- **negative** indicates project is behind schedule
- **positive** means ...

Task	Estimated Duration (month)	Estimated Cost (\$)	Status	EV
A	2	70,000	100%	
B	3	27,000	85%	
C	1.5	82,000	62%	
D	2.5	25,000	48%	
E	2	30,000	23%	
F	0.5	48,000	16%	
G	1.5	100,000	0	
H	2	10,000	0	
I	1	45,000	0	
J	2.5	17,000	0	
K	3	32,000	0	
L	1.5	12,000	0	
M	2	10,000	0	
N	3	24,000	0	
O	1	12,000	0	
Total	29	544,000	0	

PV (value of work planned to be completed)

EV = actual value of work completed
= **estimated cost** * **status** of completion



AC (actual cost)

Exercise #1

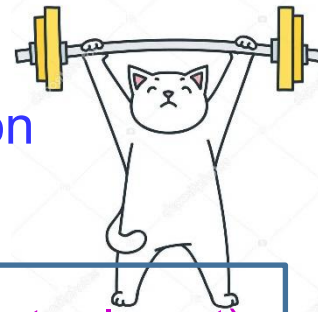
You are now in the 9th month of 29 months project. As of today, you have spent **RM198,000** based on the **invoice reconciliation**. You need to provide data to Mr. Jordon (Project Sponsor), specifying if the project status is healthy or unhealthy. You are using EVM/EVA technique to provide the sponsor with this information.

BAC (budget at completion)
i.e. project budget

Task	Estimated Duration (month)	Estimated Cost (\$)	Status	EV
A	2	70,000	100%	
B	3	27,000	85%	
C	1.5	82,000	62%	
D	2.5	25,000	48%	
E	2	30,000	23%	
F	0.5	48,000	16%	
G	1.5	100,000	0	
H	2	10,000	0	
I	1	45,000	0	
J	2.5	17,000	0	
K	3	32,000	0	
L	1.5	12,000	0	
M	2	10,000	0	
N	3	24,000	0	
O	1	12,000	0	
Total	29	544,000	0	

PV (value of work planned to be completed)

EV = actual value of work completed
= **estimated cost** * **status** of completion



Exercise #1(R)

AC (actual cost)

You are now in the 9th month of 29 months project. As of today, you have spent **RM198,000** based on the invoice reconciliation. You need to provide data to Mr. Jordon (Project Sponsor), specifying if the project status is healthy or unhealthy. You are using EVM/EVA technique to provide the sponsor with this information.

BAC (budget at completion)
i.e. project budget

Exercise #1

You are now in the 9th month of 29 months project. As of today, you have spent RM198000 based on the invoices reconciliation. You need to provide data to Mr. Jordon (Project Sponsor), specifying if the project status is healthy or unhealthy. You are using EVM/EVA technique to provide the sponsor with this information.

Answer

Task	Estimated Duration (month)	Estimated Cost (\$)	Status	EV
A	2	70,000	100%	70,000
B	3	27,000	85%	22,950
C	1.5	82,000	62%	50,840
D	2.5	25,000	48%	12,000
E	2	30,000	23%	6900
F	0.5	48,000	16%	7680
G	1.5	100,000	0	
H	2	10,000		
I	1	45,000		
J	2.5	17,000		
K	3	32,000		
L	1.5	12,000		
M	2	10,000		
N	3	24,000		
O	1	12,000		
Total	29	544,000	0	

Actual value of work completed in the 9th month

You are now in the 9th month,

AC (actual cost) = 198,000

Budget At Completion (BAC) = 544,000

PV = 70k + 27k + 82k + 25k = 204,000 (9 months)

EV = (70k*100%) + (27k*85%) + (82k*62%) + (25k*48%) + (30k*23%) + (48k*16%) = 170,370

SV (schedule variance) = EV - PV

= 170,370 - 204,000
= -33,630

value of work planned to be completed in the 9th month

SV is negative means that the project is behind schedule (In the 9th month, actual value of work completed is < value of work planned to be completed)

Exercise #2 (using different set of data)

You are now in the 9th month of 29 months project. As of today, you have spent RM198,000 based on the invoices reconciliation. You need to provide data to Mr. Jordon (Project Sponsor), specifying if the project status is healthy or unhealthy. You are using EVM/EVA technique to provide the sponsor with this information.

Task	Estimated Duration (month)	Estimated Cost (\$)	Status	EV
A	2 PV	70,000	100%	
B	3	27,000	100%	
C	1.5	82,000	100%	
D	2.5	25,000	100%	
E	2	30,000	50%	
F	0.5	48,000	50%	
G	1.5	100,000	0	
H	2	10,000		
I	1	45,000		
J	2.5	17,000		
K	3	32,000	0	
L	1.5	12,000	0	
M	2	10,000	0	
N	3	24,000	0	
O	1	12,000	0	
Total	29	544,000	0	

Actual value of work completed in the 9th month

You are now in the 9th month,
AC (actual cost) = 198,000
Budget At Completion (BAC) = 544,000

PV = 70k + 27k + 82k + 25k = **204,000** (9 months)

EV = **243,000**

SV (schedule variance) = EV - PV
 = 243,000 - 204,000
 = **39,000**

value of work **Planned** to be completed in the 9th month

- SV is positive means that the project is **ahead of schedule** (In the 9th month, **actual value of works completed is > value of works planned to be completed**)

- **MORE** works have been completed than planned!

This good news or bad news? _

Exercise #2 (using different set of data) (con't)

You are now in the 9th month of 29 months project. As of today, you have spent RM198,000 based on the invoices reconciliation. You need to provide data to Mr. Jordon (Project Sponsor), specifying if the project status is healthy or unhealthy. You are using EVM/EVA technique to provide the sponsor with this information.

Task	Estimated Duration (month)	Estimated Cost (\$)	Status	EV
A	2	70,000	100%	
B	3	27,000	100%	
C	1.5	82,000	100%	
D	2.5	25,000	100%	
E	2	30,000	50%	
F	0.5	48,000	50%	
G	1.5	100,000	0	
H	2	10,000	0	
I	1	45,000	0	
J	2.5	17,000	0	
K	3	32,000	0	
L	1.5	12,000	0	
M	2	10,000	0	
N	3	24,000	0	
O	1	12,000	0	
Total	29	544,000	0	

You are now in the 9th month,
AC (actual cost) = 198,000

$$PV = 70k + 27k + 82k + 25k = 204,000 \text{ (9 months)}$$

$$EV = 243,000$$

$$SV = 39,000$$

$$CV \text{ (cost variance)} = EV - AC$$

$$= 243,000 - 198,000$$

$$= 45,000$$

Actual amount paid

Actual value of work completed

- **CV is positive**, it means that the project is under budget.

- In the 9th month, **actual value of work completed is > amount paid**

- However, if **CV is negative**, it means that **amount paid is > actual value of work completed!**

Exercise #3 (again using different set of data)

~~You are now in the 9th month of 29 months project. As of today, you have spent RM198,000 based on the invoices reconciliation. You need to provide data to Mr. Jordon (Project Sponsor), specifying if the project status is healthy or unhealthy. You are using EVM/EVA technique to provide the sponsor with this information.~~

Task	Estimated Duration (month)	Estimated Cost (\$)	Status	EV
A	2	70,000	100%	70,000
B	3	27,000	100%	27,000
C	1.5	82,000	100%	82,000
D	2.5	25,000	100%	25,000
E	2	30,000	100%	30,000
F	0.5	48,000	100%	48,000
G	1.5	100,000	100%	100,000
H	2	10,000	100%	10,000
I	1	45,000	100%	45,000
J	2.5	17,000	100%	17,000
K	3	32,000	100%	32,000
L	1.5	12,000	100%	12,000
M	2	10,000	100%	10,000
N	3	24,000	100%	24,000
O	1	12,000	0%	
Total	29	544,000	0	

* You are now in the **28th month**

* AC (actual amt paid) = **544,000**

$$PV = 70k + 27k + \dots + 24k = 532,000$$

$$EV = 70k + 27k + \dots + 24k = 532,000$$

$$\begin{aligned}
 CV \text{ (cost variance)} &= EV - AC \\
 &= 532,000 - 544,000 \\
 &= -12,000
 \end{aligned}$$

Actual value of work completed

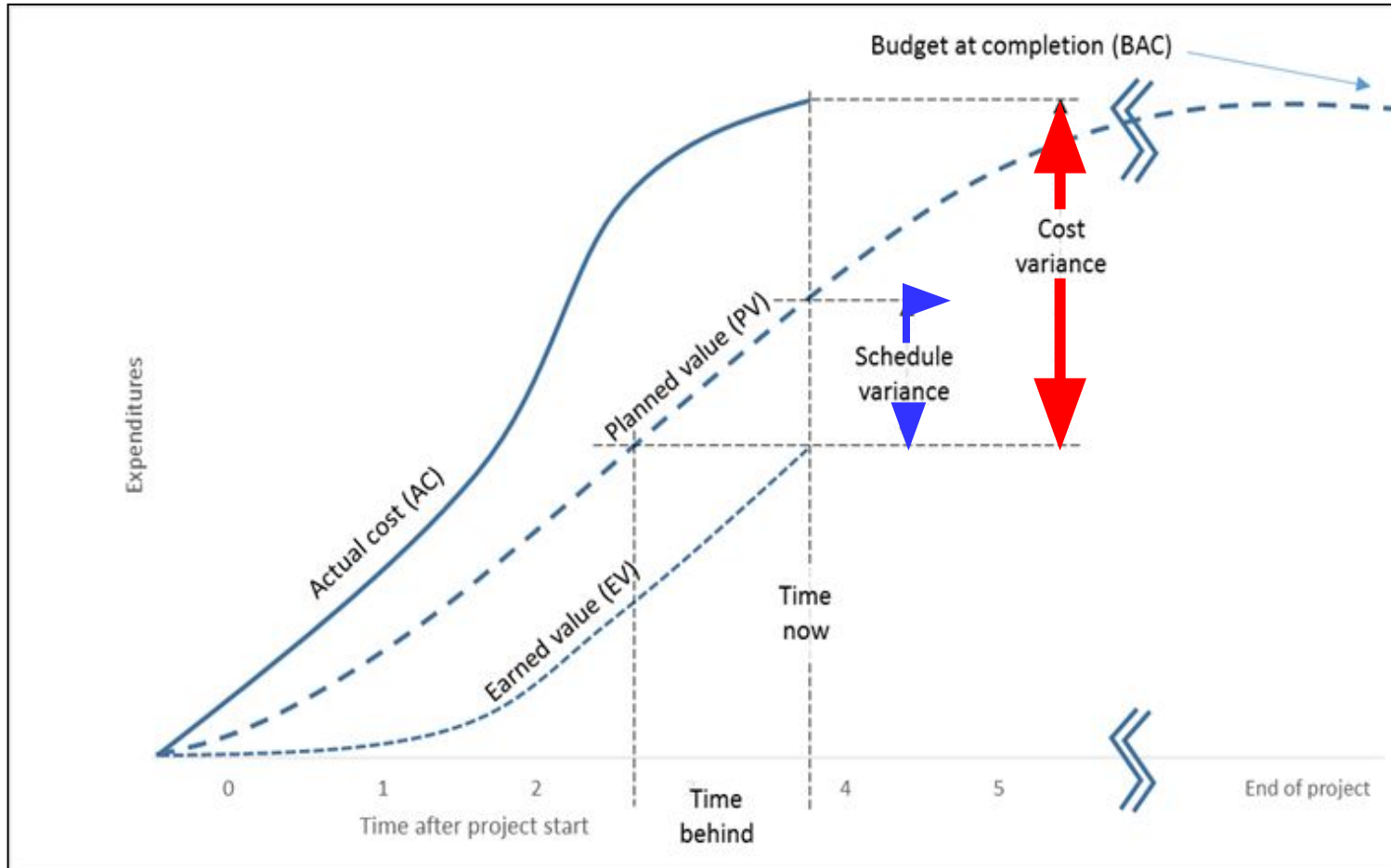
Actual amount paid

- **CV is negative**, it means that *amount paid is > actual value of work completed!*
- On the 28th month, *amount paid is more than actual value of work completed (544K vs 532K)*

Q) How much more are needed to pay complete the project? 12K

Total amount to complete the project will be **556K**

Cost Variance & Schedule Variance



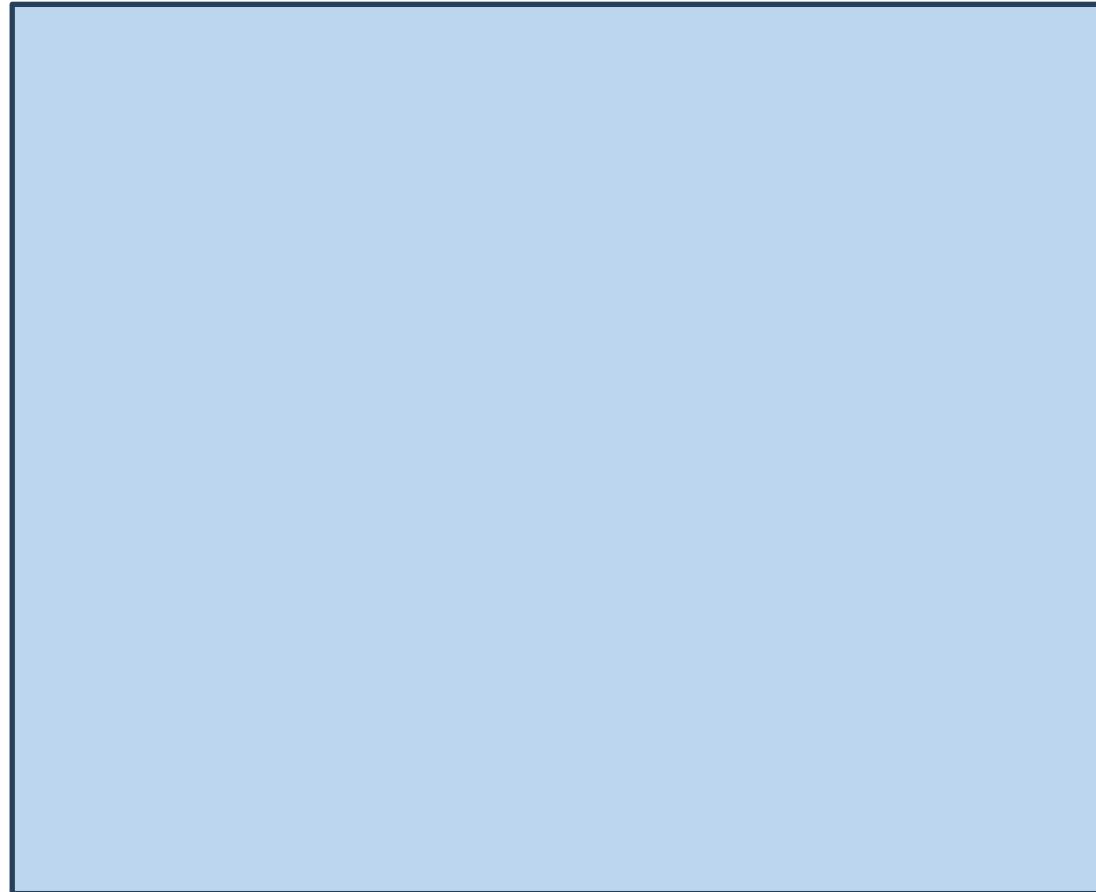
Earned Value Management (EVM)
Variances

Cost Variance (CV)
 $= EV - AC$ (negative indicates
 budget is exceeded)
*(at time t, amount paid is > actual value of
 work completed)*

Schedule Variance (SV)
 $= EV - PV$ (negative indicates
 behind schedule)
*(At time t, actual value of work completed is <
 value of work planned to be completed)*

Tut 2

You are managing a project with 12th month's durations, and now you are in 10th month. The planned value (PV) for the 10th month is RM148,000. As of today, the total amount has been paid out is RM145.000 with 80% of the works **completed**. The budget at completion (BAC) for this project is RM 180,000. Calculate **SV** = EV - PV and **CV** = EV - AC.



PV = 148K (*value of work planned to be completed in the 10th month*)

AC = 145K (*total amount paid in the 10th month, amount of work completed = 80%*)

EV = 80% of works completed x BAC
= 0.8 x 180K = 144K

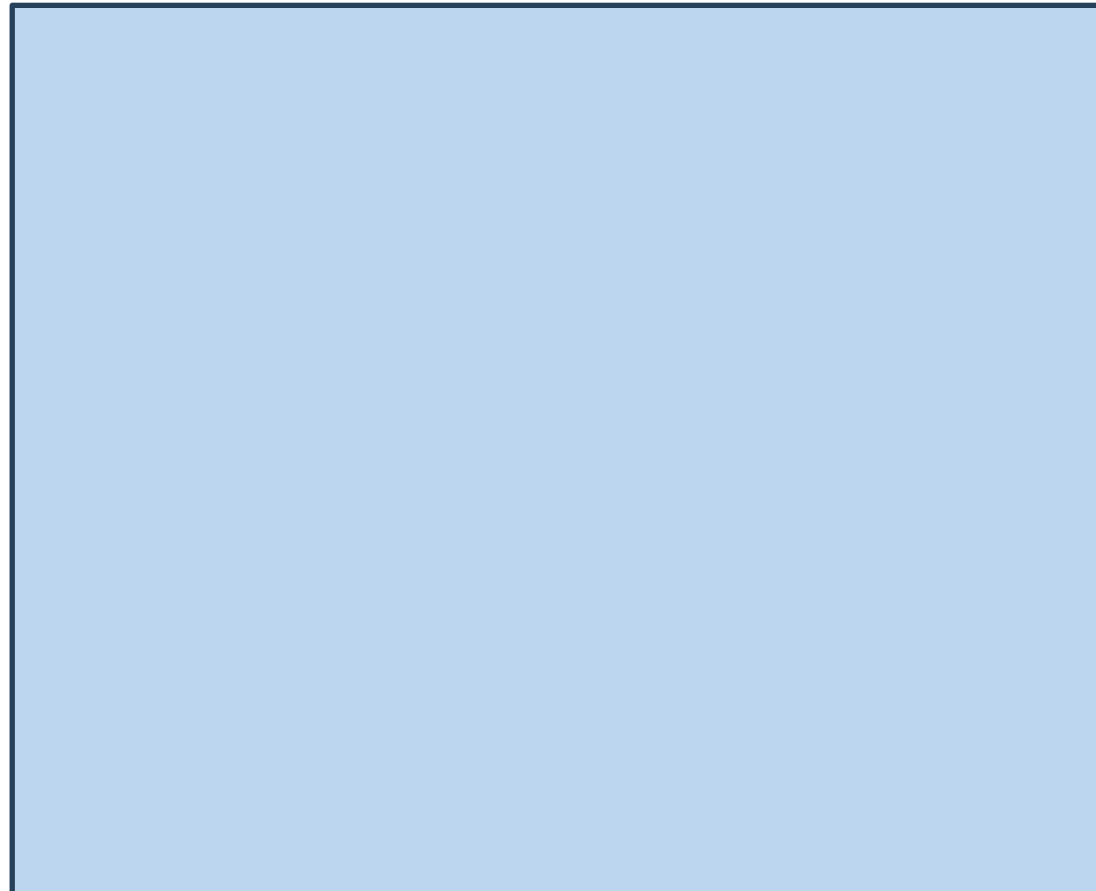
SV = 144K - 148K = -4K

CV = 144K - 145K = -1K

Q) Project progress is good? __

Tut 2b

You are managing a project with 12th month's durations, and now you are in 10th month. The planned value (PV) for the 10th month is RM148,000. As of today, the total amount has been paid out is RM145,000 **with 90% of the works completed.** The budget at completion (BAC) for this project is RM 180,000. Calculate **SV** = **EV** - **PV** and **CV** = **EV** - **AC**.



PV = 148K (*value of work planned to be completed in the 10th month*)

AC = 145K (*total amount paid in the 10th month, amount of work completed = 90%*)

EV = 90% of works completed x BAC
= 0.9 x 180K = 162K

SV = 162K - 148K = 14K

CV = 162K - 145K = 17K

Project progress is good or bad? __

Earned Value Management/ Earned Value analysis

Schedule Variance (SV) = EV – PV

- negative indicates project is behind schedule (*at time t, actual value of work completed is < value of work planned to be completed*)

- Positive indicates ?

Cost Variance (CV) = EV – AC

- negative indicates budget is exceeded (*at time t, amount paid > actual value of work completed*)

- Positive indicates ?

Select the right corrective actions to be taken when **SV** is negative:

- a. Adding more staff
- b. Adding different skills
- c. Reassigning tasks
- d. Increasing/decreasing individual supervision
- e. Improving methods of working
- f. Streamlining the process
- g. Changing resource priorities
- h. Re-planning the project
- i. Changing the phasing of deliverables
- j. Increasing/decreasing the number of inspection
- k. Encouraging the team
- l. Introducing incentives
- m. Subcontracting part of the work
- n. Negotiating changes in the specification

Earned Value Management/ Earned Value analysis

Schedule Variance (SV) = EV – PV

- negative indicates project is behind schedule (*at time t, actual value of work completed is < value of work planned to be completed*)

- Positive indicates ?

Cost Variance (CV) = EV – AC

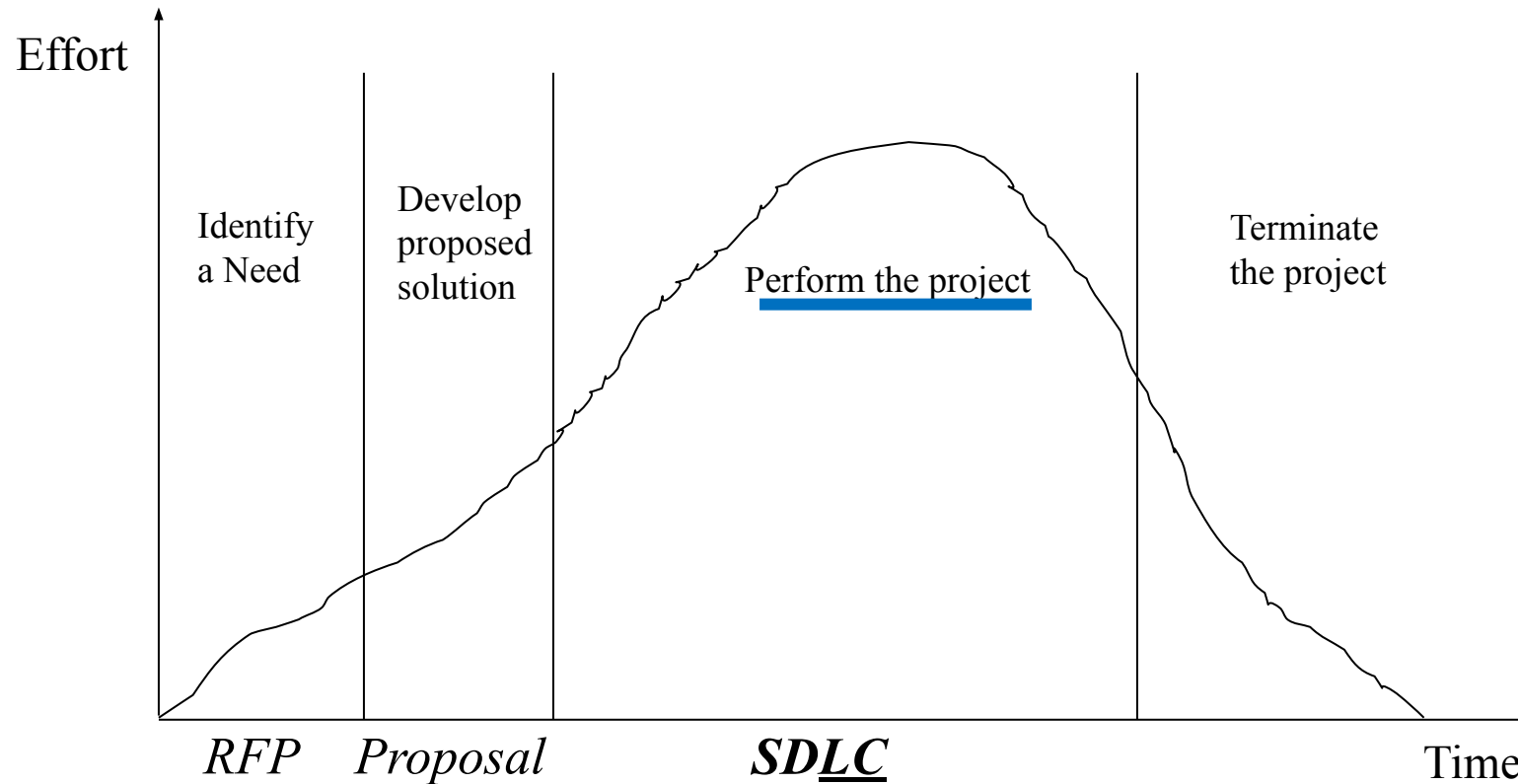
- negative indicates budget is exceeded (*at time t, amount paid > actual value of work completed*)

- Positive indicates ?

Select the right corrective actions to be taken when **CV** is negative:

- a. Adding more staff
- b. Adding different skills
- c. Reassigning tasks
- d. Increasing/decreasing individual supervision
- e. Improving methods of working
- f. Streamlining the process
- g. Changing resource priorities
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- n. Negotiating changes in the specification

2.3 SELECTION OF PROCESS MODELS

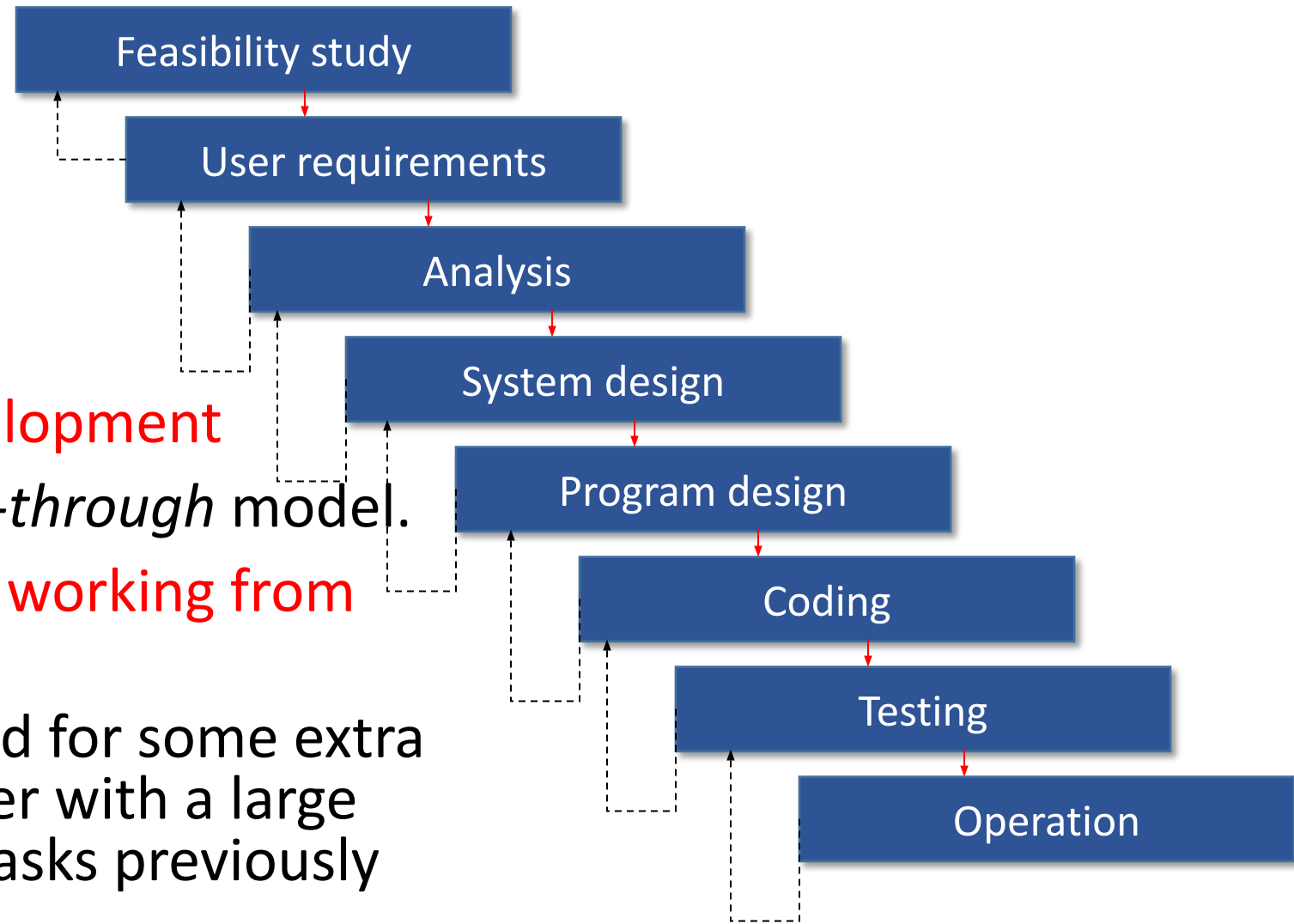


- Waterfall models
- Prototyping Model
- Rapid Application Development (RAD)
- Incremental Model
- Spiral model
- Component-based development

Waterfall Model

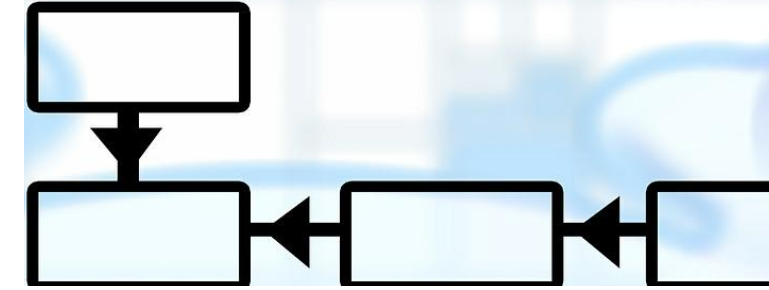
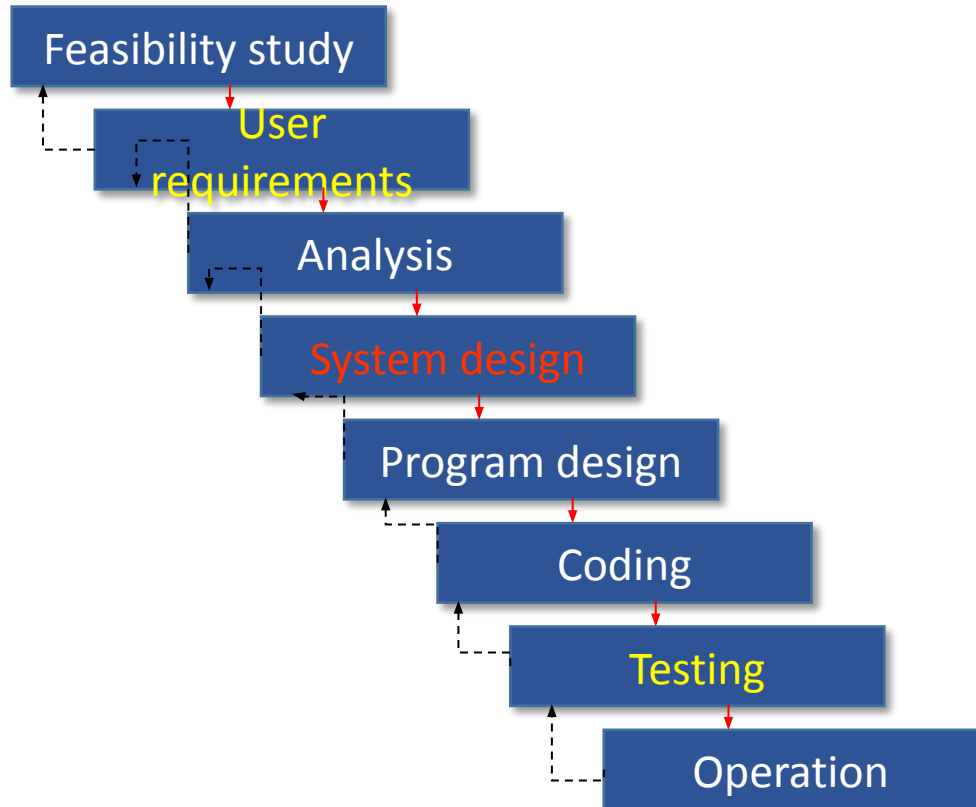
2.3 SELECTION OF PROCESS MODELS

- **‘Classical’** model of system development
- Also known as *one-shot* or *once-through* model.
- There is a **sequence of activities working from top to bottom.**
- A later stage may reveal the need for some extra work at an earlier stage. However with a large project, try to avoid reworking tasks previously thought to be completed.



Waterfall Model

2.3 SELECTION OF PROCESS MODELS



• Advantages:

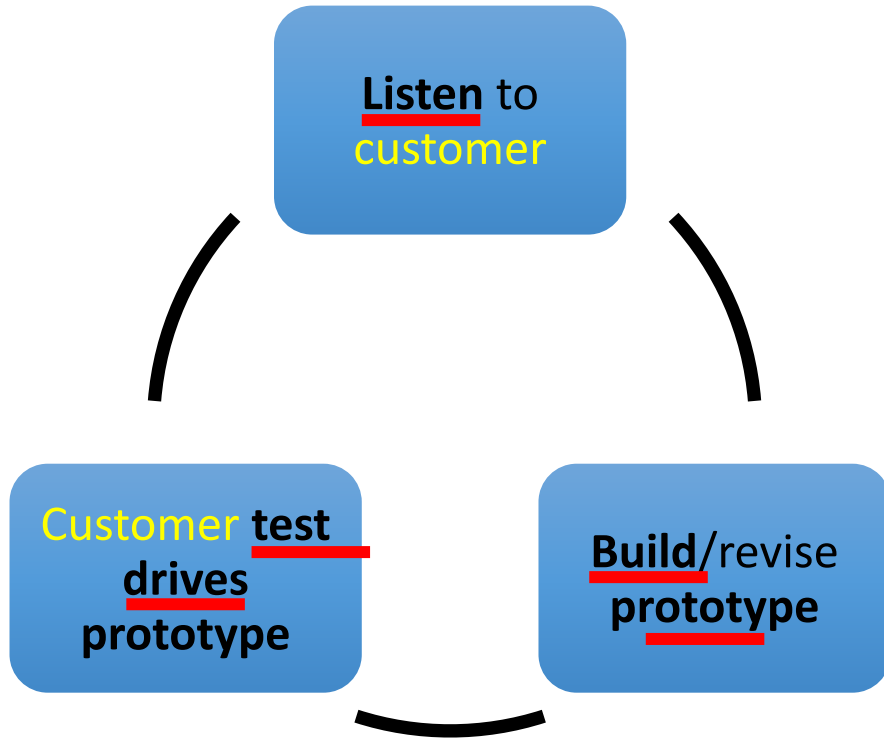
- Creates natural milestones at the end of each phase.
- Can easily check on **project progress**.
- Suitable for project where **requirements are well defined** and the development methods are well understood.

• Disadvantages:

- Not suitable for project with **uncertainty**.
(not recommended when user requirements are not well defined)
- **Not flexible** – reluctant to go back to previous stage.

Prototyping Model

2.3 SELECTION OF PROCESS MODELS



- **Evolutionary prototype**

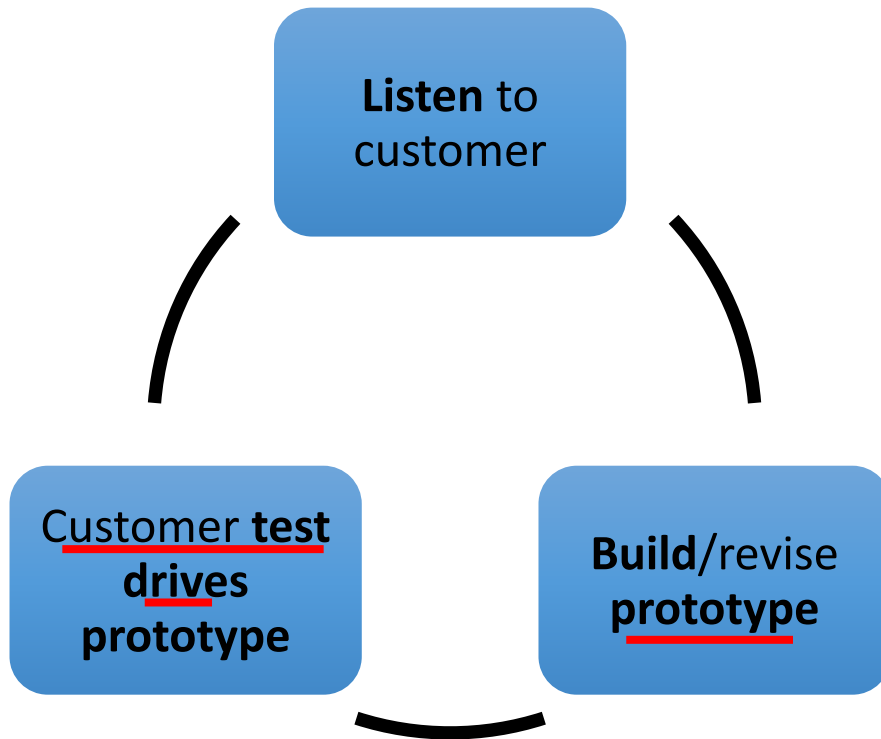
- **The prototype** is built in stages, with each stage adding more features and refinements.
- It **will evolve into the final system**

- **Throw-away prototype**

- **The prototype** is built rapidly to explore ideas and gather feedback.
- It is **discarded after the necessary information is gathered**; it is not used in the final system.

Prototyping Model

2.3 SELECTION OF PROCESS MODELS



• Advantages:

- Useful when **user requirements are uncertain**.
- Functional/valuable in designing system's UI (data-entry screen, reports or Web pages).
- Encourages **user's(customer's) involvement**.

• Disadvantages:

- Prototyping **may skip essential steps in system development** e.g. skip the testing phase in system development
- Prototyping **can be time-consuming if several iterations are needed** to refine the design. This can lead to longer project timelines.

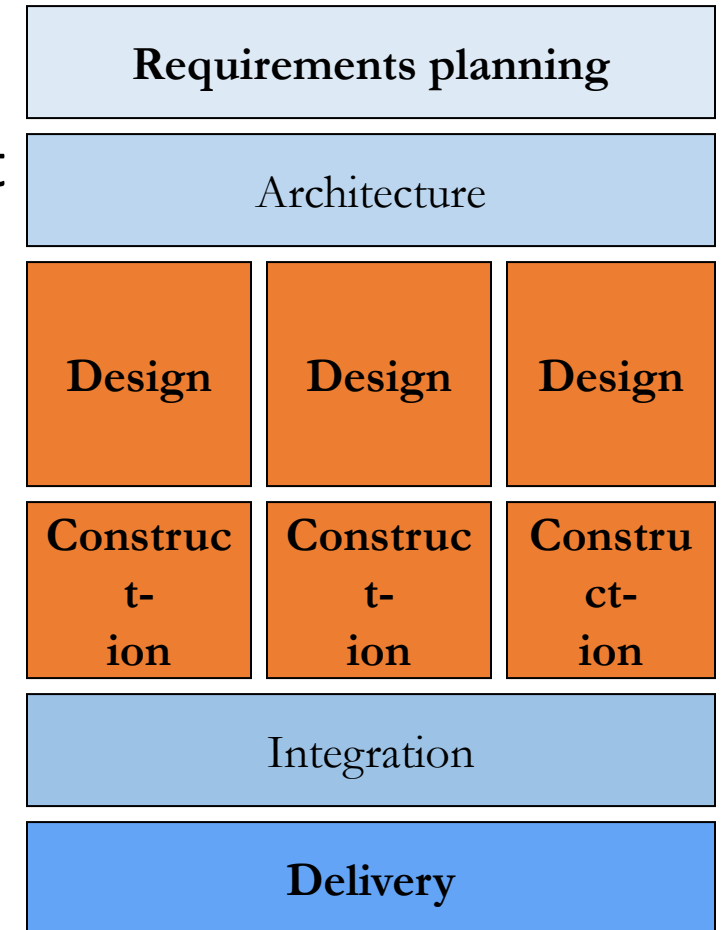
Q) Worth the time spent ? ____



Rapid Application Development (RAD)

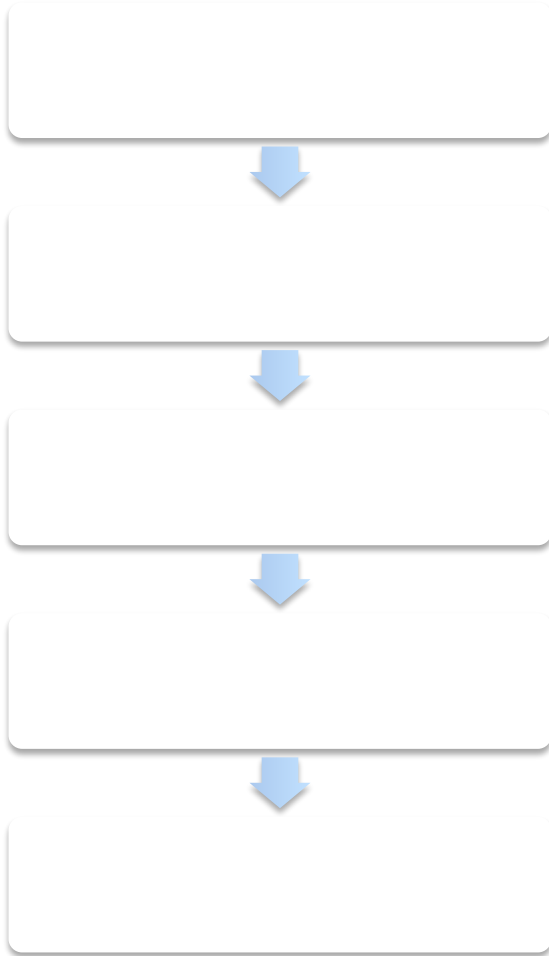
2.3 SELECTION OF PROCESS MODELS

- Incremental software development process model that **emphasizes short development time**.
- System can be produced within **60-90 days**, if requirements and project scope are well defined
- **Each RAD team will take a major function/component/module** (component to be accomplished within 3 months) and **then integrated to form a whole**.



Rapid Application Development

2.3 SELECTION OF PROCESS MODELS



- Business modeling:
 - What information drives the business process?
 - What information is generated?
 - Who generates it? Where does the information go?
 - Who processes it?
- Data modeling:
 - The information flow is refined into a set of data objects with attributes and relationships between objects.
- Process modeling:
 - Processing descriptions are created for adding, modifying, deleting, or retrieving a data object.
- Application generation:
 - Reuse existing program components/create reusable components.
- Testing and turnover:
 - Reused components have been tested before and thus reduce testing time.
 - Need to test integration of different components

Rapid Application Development

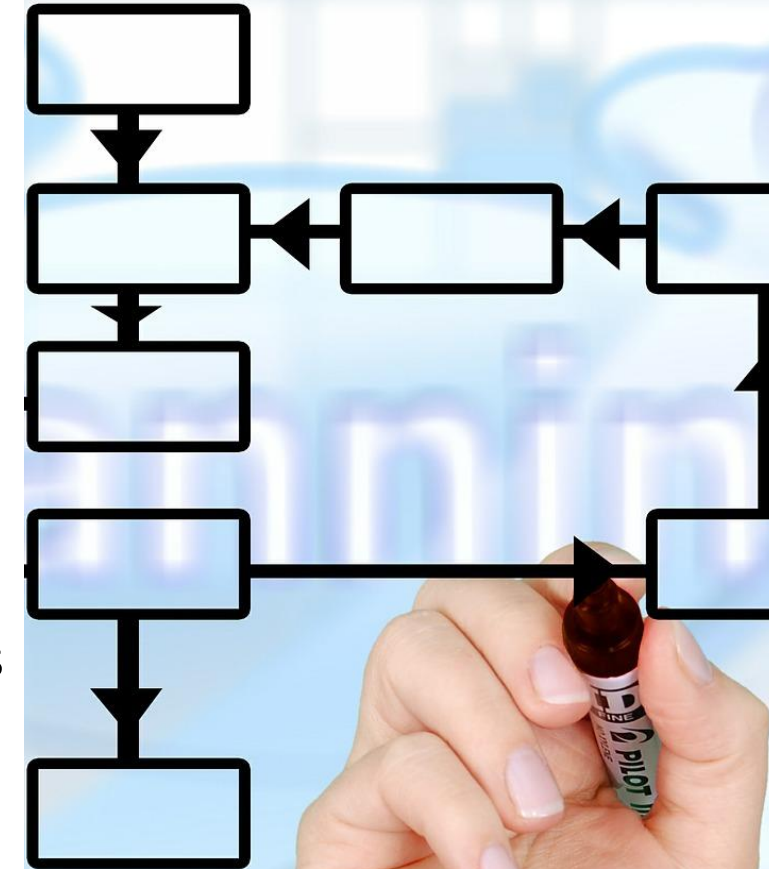
2.3 SELECTION OF PROCESS MODELS

- **Advantage**

System can be completed in a short period of time (within **60-90 days**) because several RAD teams are developing the system simultaneously

- **Drawbacks:**

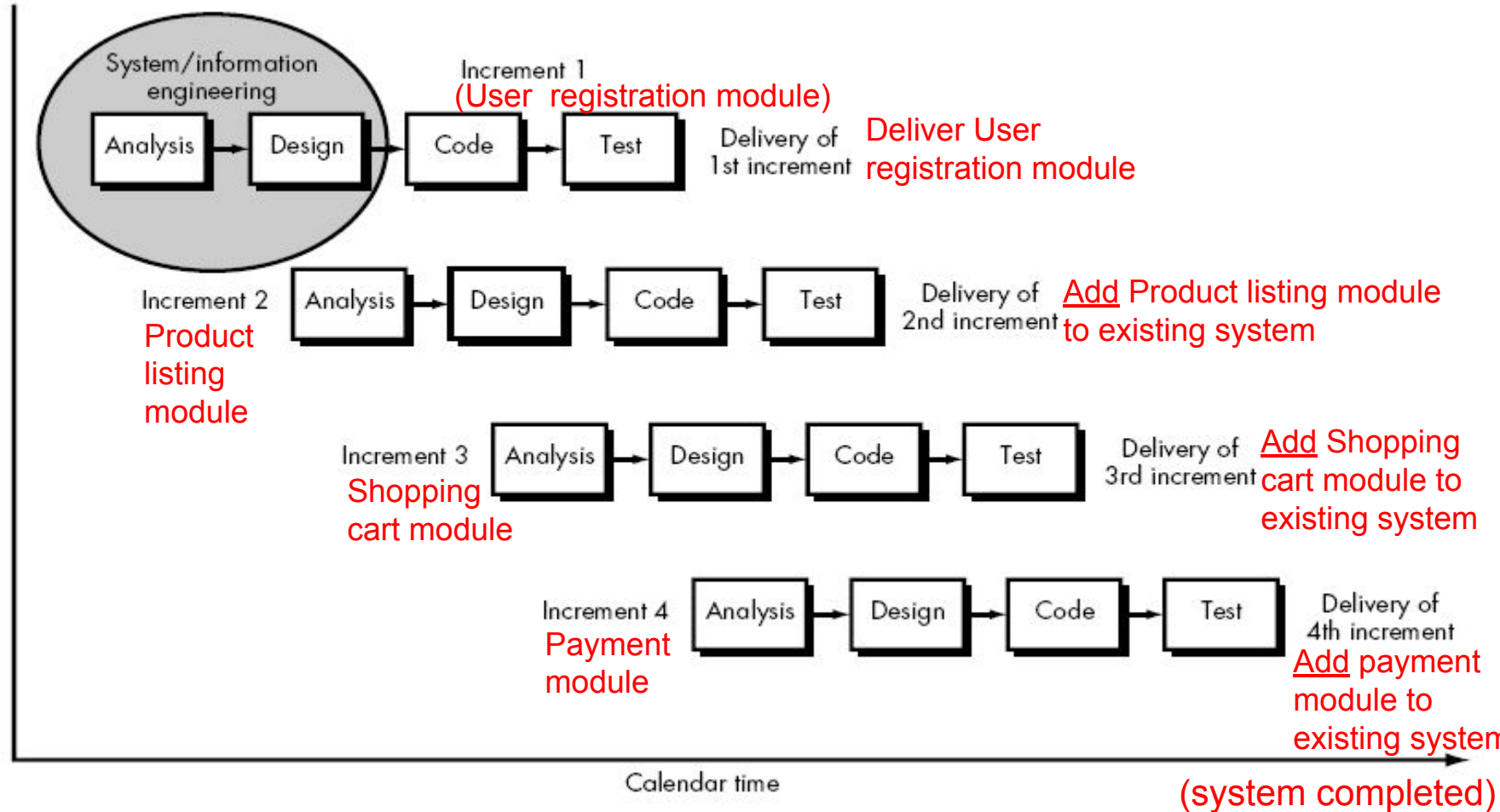
- For large but scalable projects, **RAD requires large number of resources** to create the right number of RAD teams
- Requires developers and customers committed to the rapid-fire activities.
- **Not suitable for system that cannot be modularized.**
- **Not suitable when technical risks are high: Examples**
 - New system use new technology
 - New software needs to integrate with existing system.



Incremental Model (e.g)

2.3 SELECTION OF PROCESS MODELS

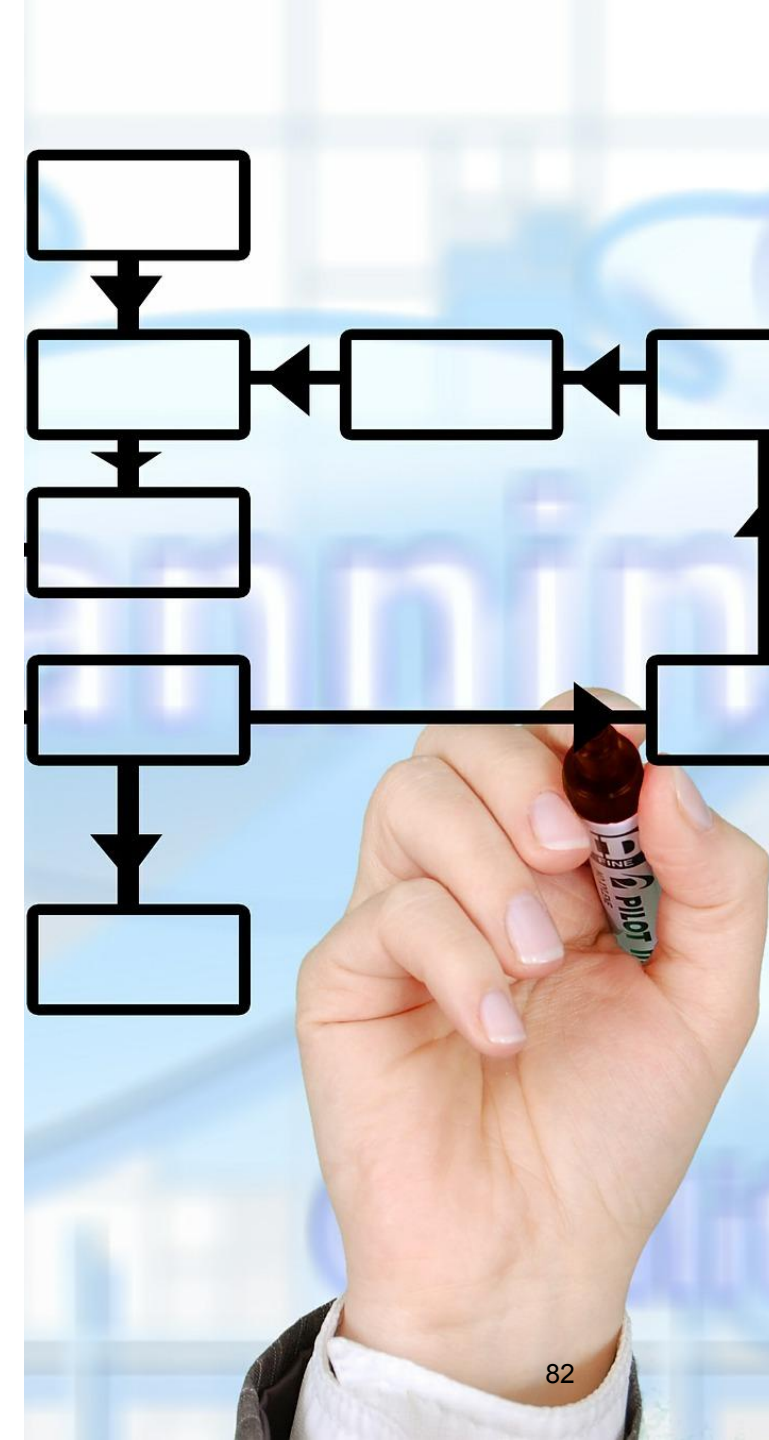
- Linear sequential model + prototyping
- The software is designed, implemented, and tested in increments (smaller portions or modules) **NOT delivering the entire system at once.**
- Each increment builds on those that have already been delivered.



Component-based Development

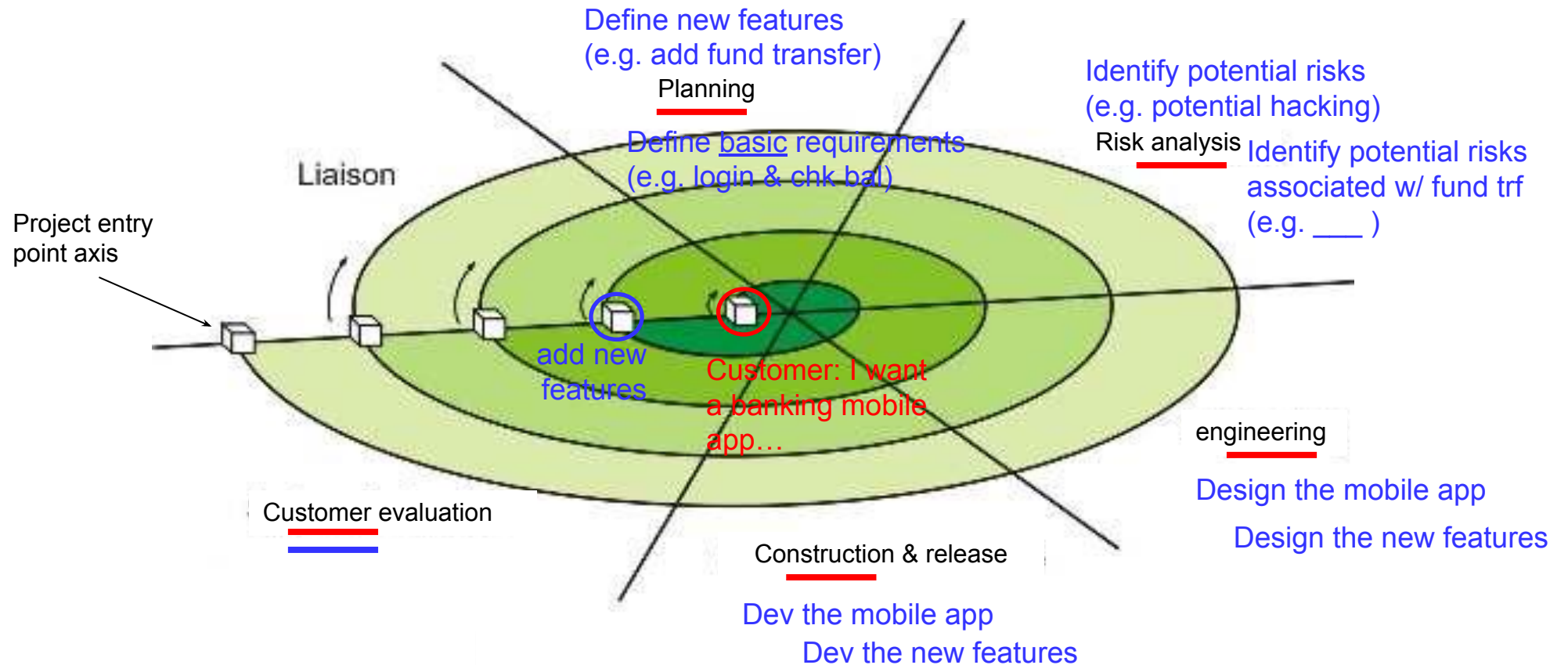
2.3 SELECTION OF PROCESS MODELS

- A software development approach that **uses well-defined independent software components**.
- These components are **then integrated to create a complete system**.
- **Example**
 - **User Registration Component:** accept registration and verify user login
 - **Search Product Component:** Allows users to search for products.
 - **Shopping Cart Component:** add, update and remove items in shopping cart.
 - **Payment Gateway Component:** Processes payments
- Using pre-built functionalities and **integrate them to form a complete system** instead of building these features from scratch



Spiral Model

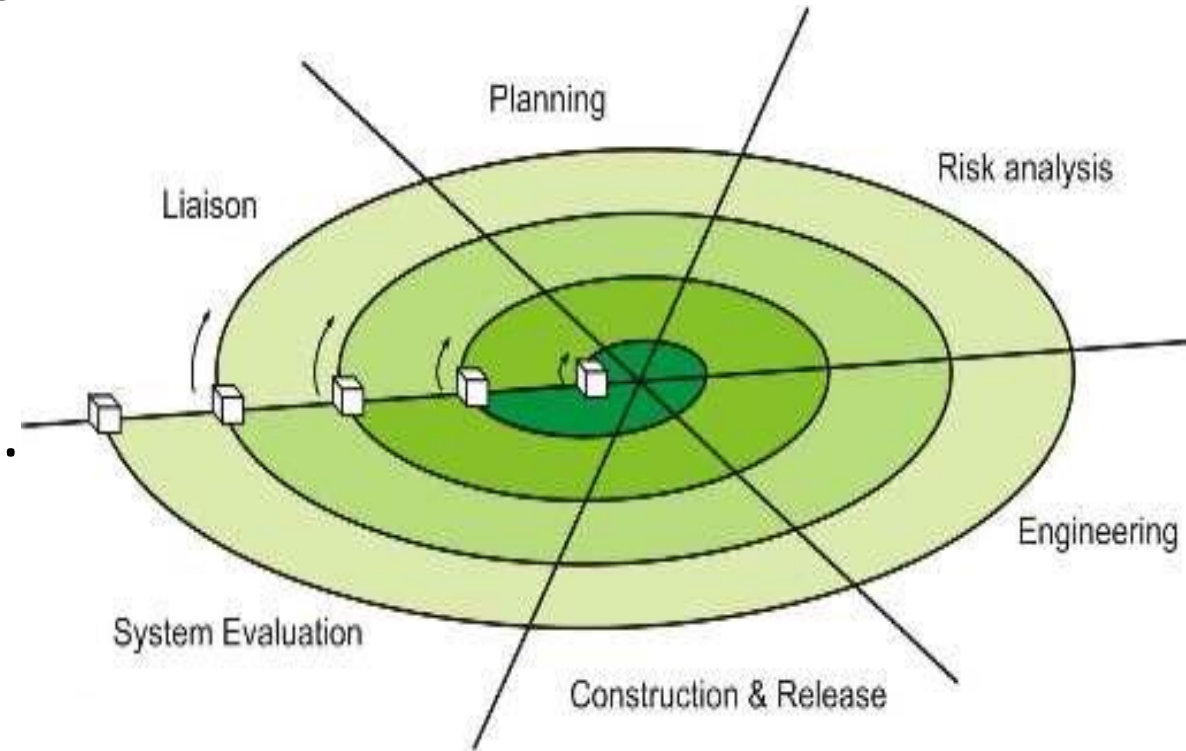
2.3 SELECTION OF PROCESS MODELS



Spiral Model

2.3 SELECTION OF PROCESS MODELS

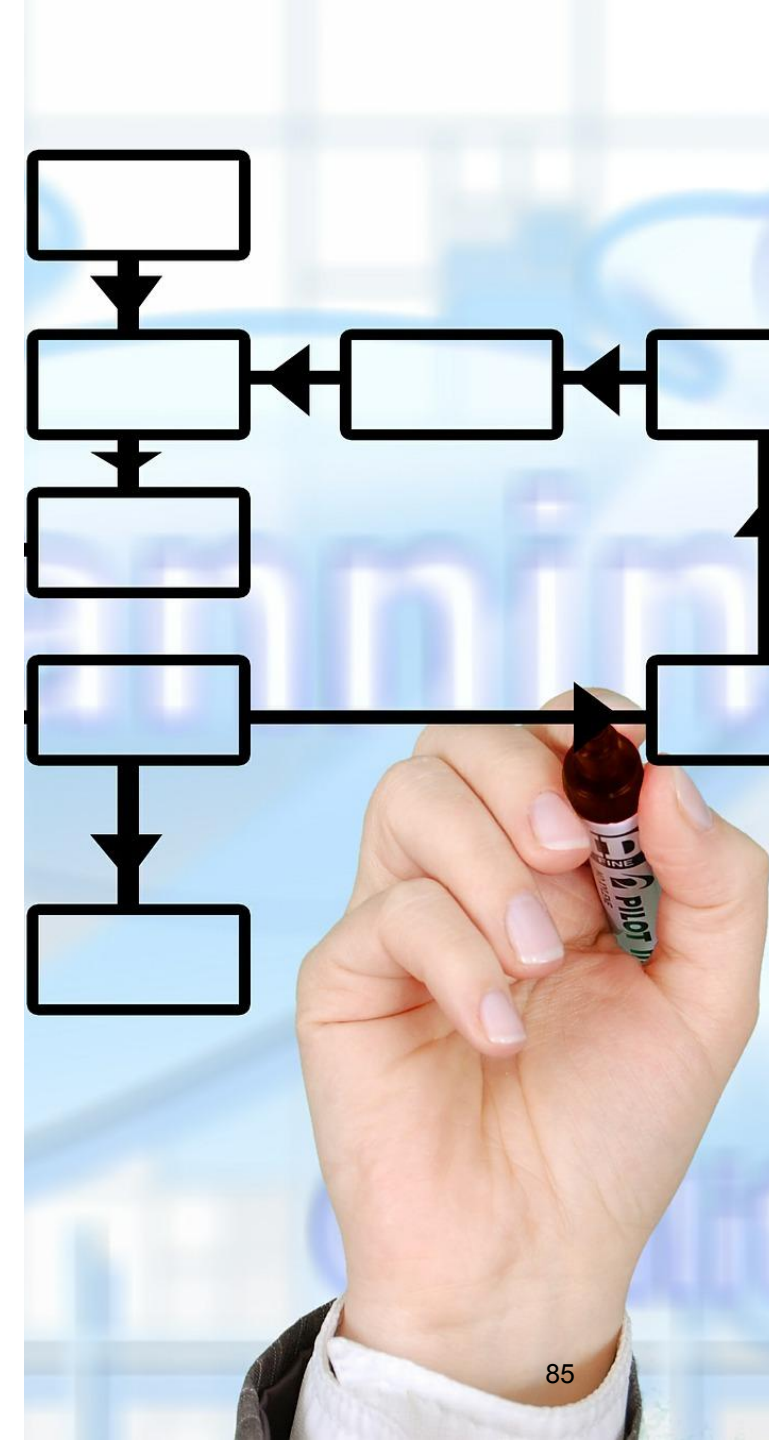
- A greater level of detail is considered at each stage of the project.
- System to be implemented is considered in more detail in each sweep in the loop.
- Each sweep terminates with an evaluation before the next iteration is embarked upon.
- **Key point: to reduce risk associated with the project**



Spiral Model

2.3 SELECTION OF PROCESS MODELS

- As this evolutionary process begins, the software engineering team moves around the spiral in a clockwise direction, beginning at the center.
- The first circuit around the spiral might result in the development of a product specification; subsequent passes around the spiral might be used to develop a prototype and then progressively more sophisticated versions of the software.
- Each pass through the planning region results in adjustments to the project plan. Cost and schedule are adjusted too.
- PM decides number of iterations required to complete the software.



Project Process Model Selection

Models	Project scope	Development Time	Budget	Man power	User Requirement	User
Waterfall	Large	<u>Long</u>	<u>High</u>	<u>Many staffs</u>	Well defined	Less involved
Prototyping	Small to medium	Flexible	<u>Low</u>	<u>Few staffs</u>	Uncertain	Highly involved
Rapid Application Development (RAD)	System can be partitioned/ modularised	60-90 days	<u>High</u>	<u>Many staffs</u> (e.g. each team to handle 1 module)	Well defined	Highly involved
Incremental	<u>Flexible</u> /complex	Short	<u>Low</u>	Lack of staff (<u>few staffs</u>)	Well defined	Highly involved
Spiral	Large-scale, risky system	Long	High	Need Experts	Uncertain	Highly involved
Component-based development	System that can be partitioned/ componentized	Short	Low	Less staffs	Well defined	Highly involved

Summary

2.1 Project Planning and
Estimation

2.2 Project Monitoring and
Control

2.3 Selection of Process
Models